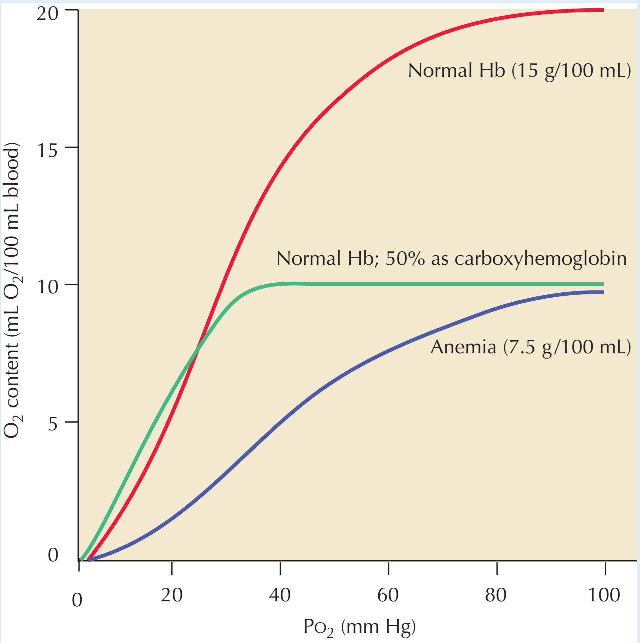
CLINICAL CORRELATE
Carbon Monoxide Poisoning

Carbon monoxide (CO) is a toxic compound produced during combustion of gasoline, propane, charcoal, natural gas, and other fuels. Prolonged exposure to air containing CO concentrations as low as 0.04% can be lethal. CO binds to hemoglobin, myoglobin, and cytochrome oxidase, producing its toxic effects. Its affinity for hemoglobin is 240 times higher than that of oxygen, and as a result, CO displaces oxygen bound to hemoglobin, forming carboxyhemoglobin and reducing oxygen-carrying capacity of hemoglobin (note the effects of a small concentration of CO on the oxyhemoglobin dissociation curve, compared with the effects of severe anemia). In addition, because binding at oxygen-binding sites of hemoglobin is cooperative, the presence of bound CO on hemoglobin increases hemoglobin affinity for oxygen (the oxyhemoglobin dissociation curve is shifted to the left), compromising dissociation of oxygen from hemoglobin and thus its delivery at normal tissue pH. Carbon monoxide poisoning is treated by respiration with 100% oxygen. Hyperbaric oxygen is also used as a treatment (patients are placed in high-pressure chambers and breathe pure oxygen), although the degree of benefit derived from this therapy is somewhat controversial. The pulse oximeter typically used in clinical settings to monitor arterial S_{O_2} relies on colorimetric measurements, but because carboxyhemoglobin, like oxyhemoglobin, is red in color, false, high S_{O_2} readings are obtained when this instrument is used in patients suffering from CO poisoning. Stated another way, the finger-probe and earlobe-probe pulse oximeters cannot differentiate between carboxyhemoglobin and oxyhemoglobin. CO poisoning results in symptoms of hypoxia, but without the typical blue appearance.



Oxygen Content of Blood: Effects of Carbon Monoxide and Anemia The arterial oxygen content of normal blood containing 15 g Hb/100 mL blood is approximately 20 mL/100 mL. In severe anemia, if hemoglobin is reduced by half, oxygen content at 100% saturation is also reduced by half. In carbon monoxide poisoning, CO binds tightly to hemoglobin, reducing the number of sites available for oxygen binding, and also shifting the oxyhemoglobin curve to the left. In the example shown above, half of the oxygen binding sites have been occupied by CO, forming carboxyhemoglobin. Thus, despite normal hemoglobin concentration, oxygen content is reduced by half in arterial blood at Pa_{O_2} of 100 mm Hg. Due to the left-shift of the dissociation curve, delivery of O_2 to tissues at normal tissue pH and P_{O_2} is greatly compromised, and thus venous P_{O_2} is profoundly reduced.

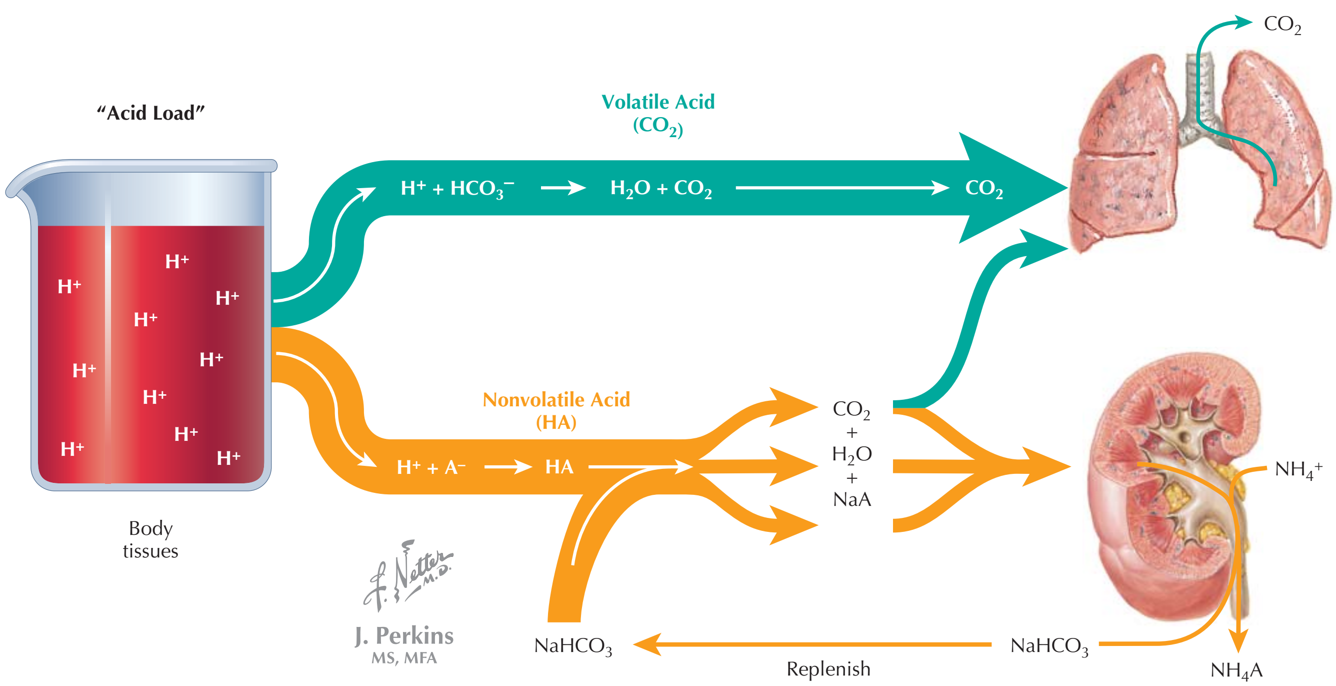


Figure 15.4 Role of Lungs and Kidneys in Acid–Base Balance The lungs and kidneys have a critical role in maintaining proper acid–base balance in the face of the acid load created by cellular metabolism of nutrients. Carbon dioxide ("volatile acid") produced by oxidative metabolism of carbohydrates and fats is efficiently eliminated by respiration in the lungs to maintain pH balance, whereas nonvolatile acids are primarily buffered by bicarbonate anion in extracellular fluid and intracellular proteins, with the kidneys replenishing the bicarbonate anion and excreting acid. H^+ is also eliminated in the urine as bicarbonate is regenerated. When a metabolic acid–base disturbance occurs, intracellular and extracellular buffering systems (involving primarily proteins and bicarbonate, respectively) are the first line of defense, along with rapid compensation by the lungs, which adjust the rate of CO_2 (volatile acid) elimination. Over a longer period (hours to days), renal mechanisms compensate by adjusting acid secretion and bicarbonate regeneration. When a respiratory acid–base disturbance occurs, the compensation is primarily renal.

important in this regard. Acid balance is also maintained by renal excretion of acid (renal mechanisms are covered in Chapter 20). Normally, CO_2 formed during metabolism of lipids and carbohydrates is readily eliminated by the respiratory system, but disturbances in respiration can result in acid–base imbalance, because changes in CO_2 elimination will directly affect carbonic acid levels. In addition, by adjusting respiration, the respiratory system can compensate for pH imbalances produced by metabolic disturbances.

The Henderson-Hasselbalch Equation

The pH of buffer systems can be calculated by the **Henderson-Hasselbalch equation:**

$$\mathrm{pH} = \mathrm{pK} + \log \frac{[A^-]}{[HA]}$$

where K is the acid dissociation constant. For the bicarbonate buffer system,

$$\mathrm{pH} = \mathrm{pK} + \log \frac{[HCO_3^-]}{[H_2CO_3]}$$

The pK for this system is 6.1. H_2CO_3 exists in low concentrations in extracellular fluids but is in equilibrium with dissolved CO_2, and thus, $0.03 \times P_{CO_2}$ (0.03 mmol/L/mm Hg is the solubility coefficient for P_{CO_2}) can be substituted for $[H_2CO_3]$ and the equation becomes:

$$\mathrm{pH} = 6.1 + \log \frac{[HCO_3^-]}{0.03 \times P_{CO_2}}$$

Substituting normal values for arterial $[HCO_3^-]$ and P_{CO_2} yields the normal pH of 7.4:

$$\mathrm{pH} = 6.1 + \log \frac{[24]}{0.03 \times 40} = 7.4$$

Based on these equations, it should be apparent that changes in P_{CO_2} will result in alteration of pH. If hypoventilation results in a rise in alveolar $P_{A_{CO_2}}$, for example, the accompanying rise in Pa_{CO_2} will cause a fall in blood pH.

ACID–BASE DISTURBANCES

Acid–base disturbances are covered briefly in the following discussion, and in detail in Chapter 20. **Acidemia** is defined as increased acidity of blood (pH below 7.35), whereas **alkalemia** is increased alkalinity of blood (pH above 7.45). Technically, **acidosis** and **alkalosis** are more general terms referring

Table 15.1 Acid–Base Disorders

Disorder	pH	1° Alteration	Defense Mechanisms
Metabolic acidosis	↓	↓ $[HCO_3^-]$	Buffers, ↓PCO_2, ↑NAE
Metabolic alkalosis	↑	↑ $[HCO_3^-]$	Buffers, ↑PCO_2, ↓NAE
Respiratory acidosis	↓	↑ PCO_2	Buffers & ↑NAE
Respiratory alkalosis	↑	↓ PCO_2	Buffers & ↓NAE

NAE, net acid excretion.
(Reprinted with permission from Hansen J: Netter's Atlas of Human Physiology, Philadelphia, Elsevier, 2002.)

to low pH and high pH, respectively, in body fluids and tissues but are usually used synonymously with acidemia and alkalemia. Alterations in pH caused by respiratory abnormality are referred to as **respiratory acidosis** or **respiratory alkalosis.** Respiratory alkalosis is caused by hyperventilation, whereas respiratory acidosis is caused by hypoventilation (due to either an acute cause, such as airway obstruction or central nervous system [CNS] depression, or a chronic lung disease). In contrast, when the primary cause of the acid–base imbalance is metabolic, for example due to a metabolic disease or abnormal renal function, it is described as **metabolic acidosis** or **metabolic alkalosis.** Compensation for respiratory acidosis or alkalosis occurs by renal mechanisms, whereas respiratory adjustments compensate for metabolic acidosis or alkalosis (Table 15.1). The general role of the kidney and lungs in acid–base balance is illustrated in Figure 15.4.

Differentiating between metabolic and respiratory causes of acidosis and alkalosis is usually a relatively simple process. To identify the condition:

- First examine the pH. Values below 7.35 are defined as acidosis; pH above 7.45 is by definition alkalosis.
- Second, examine the Pa_{CO_2}. If the pH disturbance is respiratory in origin, the Pa_{CO_2} level will be abnormal and predictive of the pH change. In other words, in respiratory acidosis, Pa_{CO_2} will be elevated above 40 mm Hg, whereas in respiratory alkalosis (caused by hyperventilation), Pa_{CO_2} will be lower than 40 mm Hg. If the condition is acute, HCO_3^- level will be normal; if the respiratory condition is chronic, HCO_3^- will be elevated in acidosis and depressed in alkalosis, reflecting renal compensation. Because of the compensation, pH will be closer to normal in chronic acidosis or alkalosis than would be expected based on a change in Pa_{CO_2} alone.
- If Pa_{CO_2} level is abnormal in the opposite direction predicted for it to be the primary alteration, the disturbance is metabolic. Examination of HCO_3^- should reveal that its level is consistent with a primary metabolic disturbance: HCO_3^- will be high in metabolic alkalosis and low in metabolic acidosis. Pa_{CO_2} levels will be depressed in metabolic acidosis and elevated in metabolic alkalosis, reflecting respiratory compensation. Because of this compensation, the pH will be closer to normal than would be predicted based on a change in HCO_3^- alone.

CONTROL OF RESPIRATION

Although breathing can be controlled voluntarily (for example, during breath holding or hyperventilation), it is ultimately an involuntary process that closely controls Pa_{O_2} and Pa_{CO_2}. Changes in both depth and rate of respiration are involved in this process. Three essential components of the involuntary control system are as follows:

- brainstem respiratory centers
- peripheral and central chemoreceptors
- mechanoreceptors in lungs and joints

Ultimately, signals involved in control of breathing are integrated in the **medullary respiratory center,** resulting in regulation of the activity of respiratory muscles (see Fig. 14.1) and affecting tidal volume and respiratory rate and pattern. Within the medulla, respiratory control is accomplished by the:

- **Ventral respiratory group,** which includes the nucleus retroambiguous, nucleus ambiguous, and nucleus retrofacialis and innervates both inspiratory and expiratory muscles. It is involved in regulation of inspiratory force and in voluntary expiration.
- **Dorsal respiratory group** within the nucleus tractus solitarius, which innervates inspiratory muscles.

The medullary respiratory center receives input from two important pontine areas:

- The **pneumotactic center,** which regulates rate and depth of respiration by cyclical inhibition of inspiration. This center has input from the cerebral cortex.
- The **apneustic center,** which stimulates inspiration. It is antagonized by the pneumotactic center.

Damage to the pons or upper medulla may result in **apneusis** (breathing characterized by prolonged inspiratory efforts and brief, intermittent exhalations). Experimentally, apneusis can be produced by ablation of the pneumotactic center and transection of the vagus nerve.

Role of Central and Peripheral Chemoreceptors

Sensory information from central and peripheral chemoreceptors is important in this regulation of respiration by the brainstem. **Central chemoreceptors** located at the ventrolateral surface of the medulla respond indirectly to changes in arterial P_{CO_2} and play a critical role in acute regulation of Pa_{CO_2}. The blood-brain barrier is largely impermeable to HCO_3^- and H^+, but CO_2 readily diffuses across the barrier and into the cerebrospinal fluid (CSF), where it affects CSF pH (by mechanisms discussed earlier). Thus, when Pa_{CO_2} is altered, respiration is affected:

- A rise in Pa_{CO_2} will cause a fall in CSF pH, which is detected by central chemoreceptors, resulting in an increase in respiratory rate.
- A fall in Pa_{CO_2} will cause a rise in CSF pH, which is detected by central chemoreceptors, resulting in decreased ventilation.

Peripheral chemoreceptors, located in the carotid bodies and aortic bodies (see Section 3), also convey information concerning the quality of arterial blood to the respiratory center in the brainstem, thereby affecting ventilation. Unlike the central chemoreceptors, these receptors respond directly to changes in Pa_{O_2} and Pa_{CO_2}, as well as pH. Through the peripheral chemoreceptor mechanism, ventilation is stimulated by the following:

- A fall in Pa_{O_2}: The ventilatory effects of changes in Pa_{O_2} are relatively small when Pa_{O_2} is above 60 mm Hg, but peripheral chemoreceptors are very responsive when Pa_{O_2} falls below this level.
- A rise in Pa_{CO_2}: Changes in Pa_{CO_2} affect respiration through both central and peripheral chemoreceptors, although the central chemoreceptor mechanism is more responsive to the changes.
- A fall in pH: Changes in H^+ concentration in arterial blood affect peripheral chemoreceptors directly, independently of the effects of Pa_{CO_2}.

Chemical control of respiration by Pa_{O_2} and Pa_{CO_2} is illustrated in Figure 15.5.

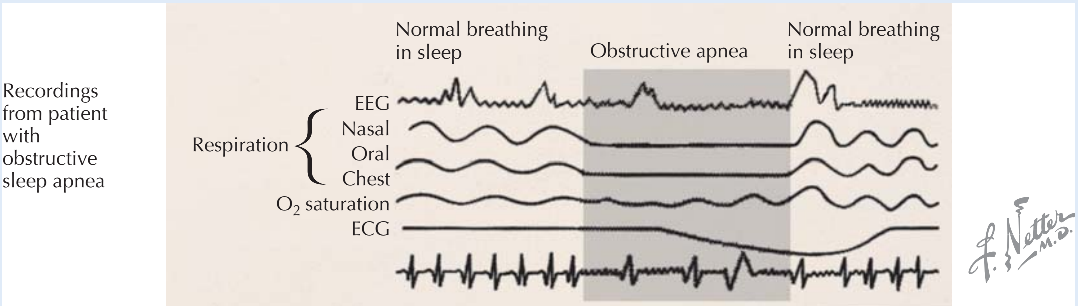
Additional Mechanisms Controlling Respiration

Respiration is also controlled by a number of additional peripheral mechanisms:

- **Pulmonary mechanoreceptors** respond to inflation of the lung and result in termination of inspiration. The afferent signals from these receptors in the smooth muscle of airway walls are transmitted through the vagus nerve to the medulla, where they inhibit the apneustic center, thereby terminating inspiration. This response to lung inflation is known as the **Hering-Breuer reflex** (specifically, the Hering-Breuer inspiratory-inhibitory reflex).
- **Irritant receptors** in the large airways respond to noxious gases and particulate matter, for example, in cigarette smoke. Activation of these receptors results in afferent

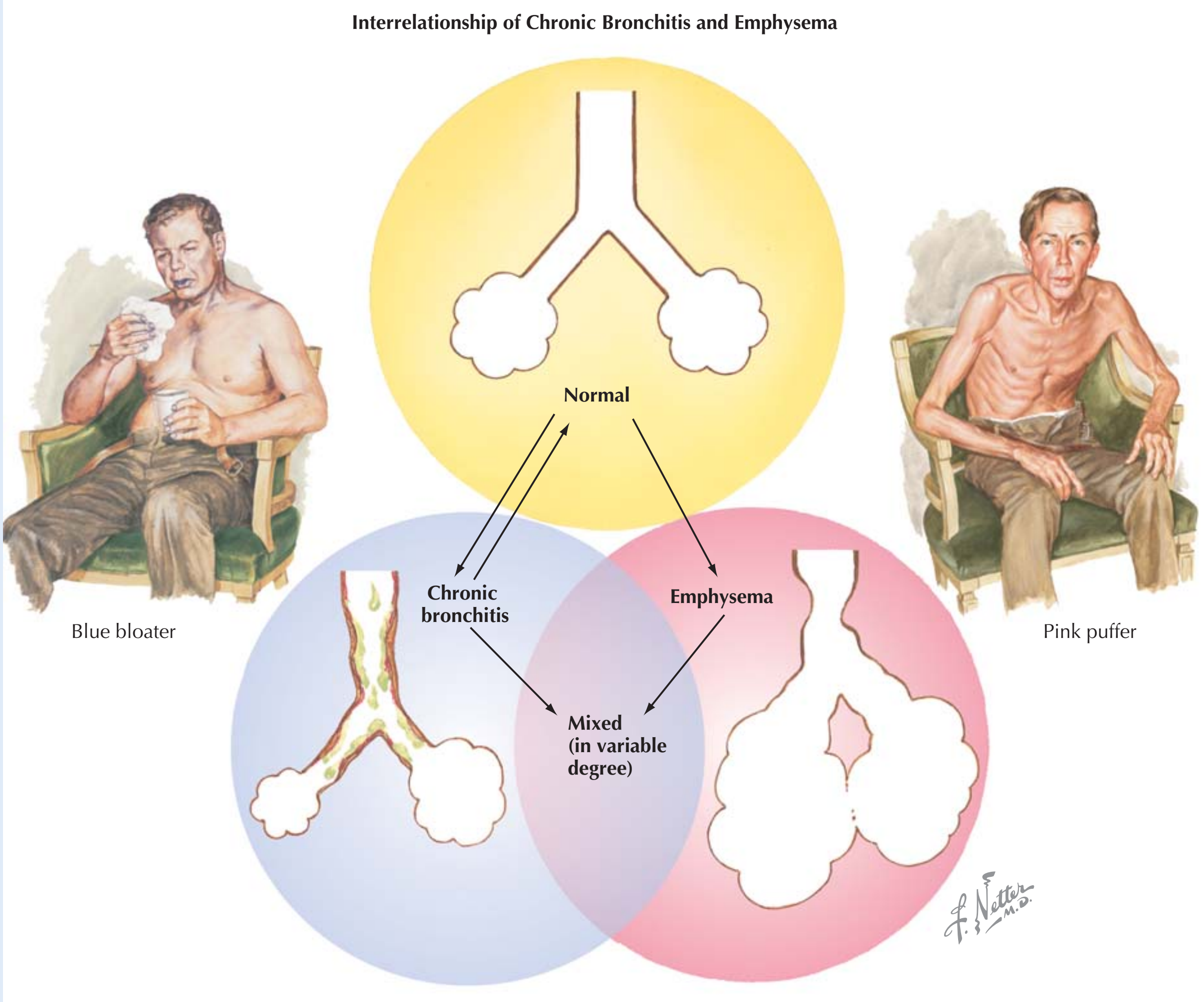
CLINICAL CORRELATE
Sleep Apnea

Sleep apnea is a disorder whereby normal breathing is periodically interrupted during sleep; the pauses in breathing can be the length of two to three breaths, and thus there can be a significant reduction in gas transport during these episodes. Sleep apnea can be central, obstructive, or complex (both central and obstructive). In central sleep apnea, brain centers are dysregulated, resulting in a lack of effort to breathe; in obstructive sleep apnea, although respiratory effort is normal, obstruction prevents airflow. Disrupted sleep and fatigue are common symptoms. Chronic sleep apnea is associated with increased incidence of heart disease and stroke.



CLINICAL CORRELATE
Respiratory Control in Chronic Obstructive Pulmonary Disease

Chronic obstructive pulmonary disease (COPD) is most often associated with tobacco smoking and may be exacerbated by occupational exposure to certain pollutants. It is characterized by chronic bronchitis (coughing with sputum production) and emphysema (destruction of alveolar walls resulting in fewer, enlarged "alveoli" with reduction in total surface area for gas diffusion). As a result, patients with this disease become hypoxic and hypercapnic (Pa_{O_2} is below normal and Pa_{CO_2} is elevated); pH is somewhat lower than normal, although significant compensation occurs by elevation of HCO_3^- through renal mechanisms. In other words, the patients suffer from chronic respiratory acidosis, with metabolic compensation. In this state of chronic hypercapnea, normal CSF pH is maintained by elevation of HCO_3^- in the CSF. The CNS is chronically exposed to high P_{CO_2}, and the central chemoreceptors become unresponsive to CO_2. Thus, patients with COPD develop "hypoxic respiratory drive," in which respiratory drive is mainly mediated by peripheral chemoreceptor responses to low Pa_{O_2}. When an acute exacerbation arises in such a patient, if supplemental O_2 is administered carelessly, results can be catastrophic and even fatal: Pa_{O_2} will rise, but respiratory drive may fail as a result, causing a fall in minute ventilation and a further rise in Pa_{CO_2}. Ventilatory support with some supplemental oxygen may be beneficial, but suppression of ventilatory drive must be avoided.



COPD is characterized by chronic bronchitis and emphysema. COPD patients usually suffer to some extent from both and are classified based on the dominant symptomatology. "Pink puffers" suffer mainly from emphysema and have ruddy complexions and elevated respiratory rate; "blue bloaters" suffer mainly from chronic bronchitis, resulting in hypoxema, cyanosis (bluish lips and skin), and often, symptoms of right heart failure, including swelling of feet and ankles ("bloating").

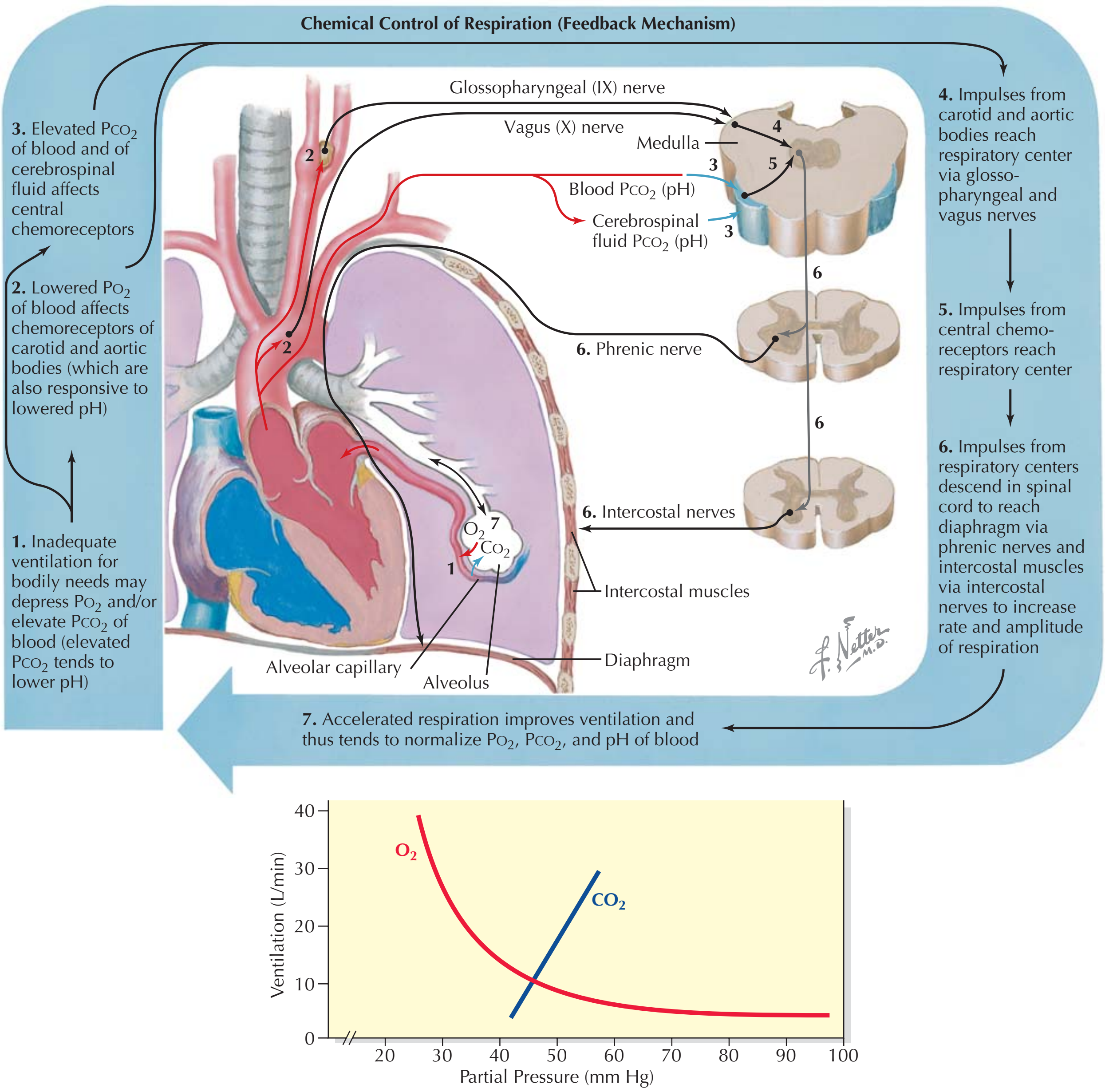


Figure 15.5 **Control of Respiration** Central and peripheral chemoreceptors regulate respiration by responding to arterial blood gas levels. Central chemoreceptors respond primarily to changes in arterial P_{CO_2}, which diffuses into the CSF and alters pH of the CSF (the blood-brain barrier is largely impermeable to HCO_3^- and H^+), while peripheral chemoreceptors in the carotid bodies and aortic bodies respond to changes in Pa_{O_2}, and also Pa_{CO_2} and pH. Brainstem respiratory centers adjust the rate and depth of respiration, producing changes in Pa_{O_2} and Pa_{CO_2} (and thus pH). In the bottom graph, the effects of Pa_{CO_2} and Pa_{O_2} on minute ventilation are illustrated.

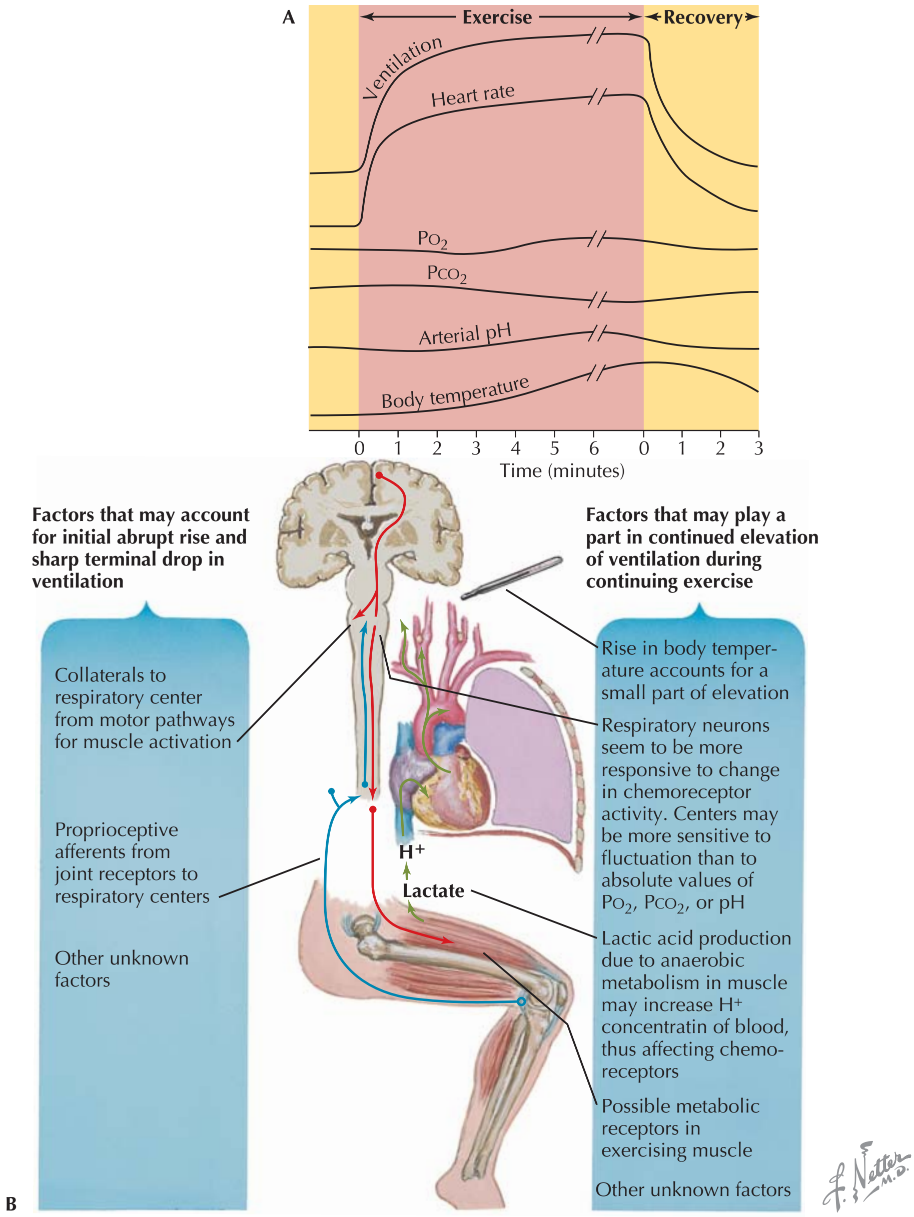


Figure 15.6 **Respiratory Response to Exercise** Increased oxygen consumption and carbon dioxide production during exercise requires adjustments of cardiac output and respiration *(A)*. Factors accounting for the rapid adjustments in respiration at the onset and termination of exercise, as well as feedback mechanisms during continued exercise, are illustrated *(B)*.

signals to the CNS mainly through the vagus nerve and causes reflexive bronchoconstriction and coughing.

- **Juxtacapillary receptors** (J receptors) in the alveoli are stimulated by hyperinflation of the lungs and various chemical stimuli; reflexive rapid, shallow breathing occurs as a result.
- **Joint and muscle mechanoreceptors** are stimulated during movement of joints and muscles, producing an increase in respiratory rate.

Respiratory Control in Exercise

Control of respiration is a critical component of the **integrated response to exercise** (Fig. 15.6). During dynamic (aerobic) exercise, oxygen consumption rises from the average resting rate of 250 mL O_2/min to as high as 4 L O_2/min, without substantial change in either Pa_{O_2} or Pa_{CO_2}. At the onset of dynamic exercise, there is a rapid increase in respiration through neural and reflexive mechanisms, although the control mechanisms are not fully understood. Activation of motor pathways results in collateral activation of the respiratory center, and respiration is further stimulated by afferent signals from muscle and joint mechanoreceptors and other, unknown factors. With continuing exercise, feedback mechanisms become important. The rise in core body temperature and elevation of lactic acid production (and plasma H^+ concentration) contribute to the further, more gradual rise in ventilation. Although Pa_{O_2} and Pa_{CO_2} change only modestly (except at high levels of exercise), respiratory control systems may be more sensitive to such changes during exercise. When exercise is terminated, ventilation diminishes rapidly at first but requires some time to fall to the resting level, due to the continued activation of feedback mechanisms until the metabolic alterations associated with exercise (including the elevation of lactic acid) are reversed.

Adaptation to High Altitude

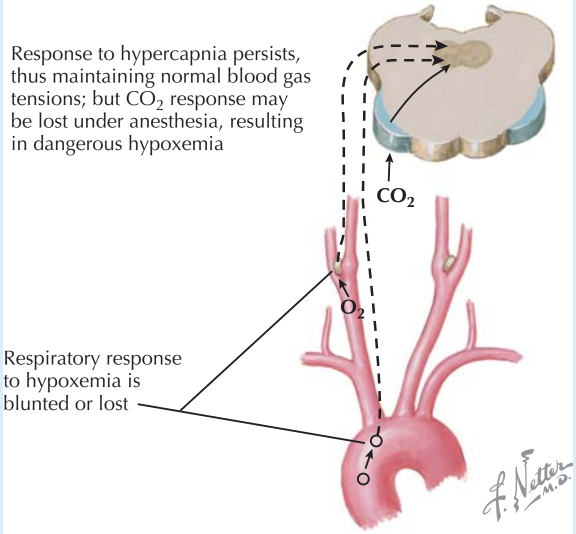
The control of respiration is also important in **adaptation to high altitude.** At the lower atmospheric pressures associated with high altitudes, the PO_2 of inspired air is reduced, resulting in hypoxemia. The fall in Pa_{O_2} stimulates ventilation through peripheral chemoreceptors, but this effect is tempered by the resulting fall in Pa_{CO_2} and the accompanying alkalosis, which inhibit ventilation through central and peripheral chemoreceptor mechanisms. Over a period of time, renal compensatory mechanisms result in elevation of plasma HCO_3^-, and as pH returns to normal, ventilation again increases. In addition to increased ventilation, other factors that contribute to the adaptation to high altitude include the following:

- Hypoxemia stimulates red blood cell production; the resulting **polycythemia** (and higher plasma hemoglobin) increases the oxygen-carrying capacity of blood.
- Elevation of **2,3-DPG** causes a rightward shift of the oxyhemoglobin dissociation curve, and thus, oxygen more readily dissociates from hemoglobin at the tissue level.

CLINICAL CORRELATE
Acute Altitude Sickness

Acute altitude sickness is a relatively common response to high altitude (greater than 8000 feet above sea level) in people who are accustomed to living at low altitudes, especially when the ascent is rapid. Symptoms include headache, tachycardia, shortness of breath, nausea and vomiting, loss of appetite, lightheadedness, and fatigue. In most cases, symptoms are mild and resolve in a matter of days as acclimation takes place. In rare, extreme cases, life-threatening pulmonary edema or cerebral edema may develop. Pulmonary edema occurs as a result of pulmonary vasoconstriction and an increase in pulmonary vascular permeability. As discussed earlier, pulmonary vasoconstriction is a normal response to alveolar hypoxia, but it is exaggerated in high-altitude pulmonary edema. The cause of high-altitude cerebral edema has not been established but likely involves cerebral vasodilation in response to hypoxemia, resulting in high capillary hydrostatic pressure. Descent to lower altitude is a critical component of treatment in these severe cases.

Effects of High Altitude on Respiratory Mechanism



Review Questions

CHAPTER 13: PULMONARY VENTILATION AND PERFUSION AND DIFFUSION OF GASES

1. The reduction in pulmonary vascular resistance that occurs when pulmonary artery pressure is increased is mainly a result of:

A. recruitment and distension of pulmonary capillaries.
B. autoregulation by myogenic mechanisms.
C. redistribution of pulmonary blood flow.
D. metabolic vasodilation.
E. active vasodilation of pulmonary arterioles.

2. Which of the following lung volumes or capacities can **NOT** be measured by spirometry alone?

A. Tidal volume
B. Expiratory reserve volume
C. Inspiratory reserve volume
D. Total lung capacity
E. Vital capacity

3. With regard to the ventilation, the perfusion, or the ventilation-to-perfusion ratio in regions of the lung in a person in the standing position,

A. perfusion is highest at the top of the lung.
B. ventilation is lowest at the bottom of the lung.
C. the ventilation-to-perfusion ratio is greatest at the top of the lung.
D. the ventilation-to-perfusion ratio approaches infinity in areas of shunt.
E. the ventilation-to-perfusion ratio is zero in areas of dead space.

4. Diffusion of gas through a membrane:

A. is inversely related to the surface area for diffusion.
B. is directly related to the difference in partial pressure of the gas on each side of the membrane.
C. requires active transport.
D. is inversely related to the diffusion constant of the gas.
E. is directly related to the thickness of the membrane.

5. In zone 2 of the lung (in a standing person),

A. both pulmonary arterial and venous pressures exceed alveolar pressure.
B. both pulmonary arterial and venous pressures are below alveolar pressure.
C. alveolar pressure is higher than pulmonary venous pressure but less than arterial pressure.
D. the ventilation-to-perfusion ratio approaches zero.
E. the ventilation-to-perfusion ratio approaches infinity.

CHAPTER 14: THE MECHANICS OF BREATHING

6. The mechanical system that produces breathing is at rest (outward elastic recoil pressure of the chest wall is equal to and opposing inward elastic recoil pressure of the lungs) at:

A. residual volume.
B. functional residual capacity.
C. 60% of total lung capacity.
D. 70% of total lung capacity.
E. total lung capacity.

7. In the respiratory system as a whole, the greatest resistance to flow occurs in the:

A. respiratory bronchioles.
B. terminal bronchioles.
C. medium-sized airways.
D. bronchi.
E. trachea.

8. Dynamic compression of airways is responsible for:

A. effort independence of expiratory flow.
B. the vital capacity of the lung.
C. normal resting tidal volume.
D. the total lung capacity that can be achieved during inspiration.
E. peak flow during expiration.

9. In severe chronic obstructive pulmonary disease (COPD),

A. lung compliance is reduced.
B. elastic recoil of the lung is decreased.
C. total lung capacity is reduced.
D. functional residual capacity is reduced.
E. residual volume is reduced.

10. Which of the following is **NOT** a characteristic or function of surfactant?

A. Decreases pulmonary compliance
B. Reduces the work of breathing
C. Reduces surface tension of alveoli and small airways
D. Deficient in respiratory distress syndrome
E. Contains dipalmitoyl phosphatidyl choline

CHAPTER 15: OXYGEN AND CARBON DIOXIDE TRANSPORT AND CONTROL OF RESPIRATION

11. Which of the following changes will result in the greatest increase in oxygen content of arterial blood, assuming normal alveolar oxygen concentration?

A. An increase in hematocrit from 40 to 45
B. An increase in alveolar oxygen concentration from 100 to 150 mm Hg
C. A 10% increase in blood level of 2,3-DPG
D. A fall in blood pH from 7.4 to 7.35
E. A 10% increase in alveolar ventilation

12. Lab tests reveal that a patient has blood pH of 7.3, elevated arterial P_{CO_2}, and slightly elevated arterial plasma bicarbonate level. The acid–base status is:

A. metabolic acidosis.
B. metabolic alkalosis.
C. respiratory acidosis.
D. respiratory alkalosis.
E. impossible to determine from these values.

13. The major mechanism of long-term adaptation to high altitude is:

A. increased heart rate.
B. increased respiratory rate.
C. reduced 2,3-DPG in blood.
D. polycythemia.
E. reduced plasma bicarbonate.

14. In the control of respiration, central chemoreceptors respond mainly to changes in:

A. arterial pH.
B. arterial P_{CO_2}.
C. 2,3-DPG in blood.
D. arterial HCO_3^-.
E. arterial O_2.

15. The rapid adjustment of respiratory rate at the onset of exercise is mediated in part by:

A. a fall in arterial pH.
B. a rise in arterial P_{CO_2}.
C. 2,3-DPG in blood.
D. mechanoreceptors in joints.
E. a rise in body temperature.

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Section 5

RENAL PHYSIOLOGY

The kidneys are the primary avenue for regulating extracellular fluid (ECF) and electrolyte homeostasis. Their "job" is to regulate proper ECF volume and solute composition on a minute-to-minute basis. This is accomplished by intrarenal physical forces and feedback systems, as well as input from the nervous and endocrine systems. At the same time, the kidneys excrete waste (excess fluid and electrolytes, as well as urea, bilirubin, drugs, and potential toxins) and provide key endocrine functions.

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CHAPTER 16

Overview, Glomerular Filtration, and Renal Clearance

STRUCTURE AND OVERALL FUNCTION OF THE KIDNEYS

The kidneys perform a host of functions, including the following:

- Regulation of fluid and electrolyte balance: The kidneys regulate the volume of extracellular fluid through reabsorption and excretion of NaCl and water. They also regulate the plasma levels of other key substances (Na^+, K^+, Cl^-, HCO_3^-, H^+, glucose, amino acids, Ca^{2+}, phosphates). Key renal processes that allow regulation of circulating substances are as follows:
  - *Filtration* of fluid and solutes from the plasma into the nephrons
  - *Reabsorption* of fluid and solutes out of the renal tubules into the peritubular capillaries
  - *Secretion* of select substances from the peritubular capillaries into the tubular fluid, which facilitates excretion of the substances; both endogenous (e.g., K^+, H^+, creatinine, ACh, NE) and exogenous (e.g., para-aminohippurate, salicylic acid, penicillin) can be secreted in the urine
  - *Excretion* of excess fluid, electrolytes, and other substances (e.g., urea, bilirubin, acid [H^+])
- Regulation of plasma osmolarity: "Opening" and "closing" specific water channels in the renal collecting ducts produces concentrated and dilute urine (respectively), allowing regulation of plasma osmolarity and extracellular fluid (ECF) volume.
- Excretion of metabolic waste products: Urea (from protein metabolism), creatinine (from muscle metabolism), bilirubin (from breakdown of hemoglobin), uric acid (from breakdown of nucleic acids), metabolic acids, and foreign substances such as drugs are eliminated in urine.
- Producing/converting hormones: The kidney produces **erythropoietin** and **renin.** Erythropoietin stimulates red blood cell production in bone marrow. **Renin,** a proteolytic enzyme, is secreted into the blood and converts angiotensinogen to angiotensin I (which is then converted to angiotensin II by angiotensin-converting enzyme [ACE] in lungs and other tissues). The renin-angiotensin system is critical for fluid–electrolyte homeostasis and long-term blood pressure regulation. The renal tubules also convert 25-hydroxyvitamin D to the active **1,25-dihydroxyvitamin D,** which can act on kidney, intestine, and bone to regulate calcium homeostasis.
- Metabolism: Renal **ammoniagenesis** has an important role in acid–base homeostasis (discussed further in Chapter 20). During starvation, the kidney also has the ability to produce glucose through **gluconeogenesis.**

The kidneys are bilateral, retroperitoneal organs that receive their blood supply from the renal arteries (Fig. 16.1*A*). Each kidney is approximately the size of an adult fist, surrounded by a fibrous capsule. The parenchyma is divided into the cortex and outer and inner medulla. The cortex contains renal corpuscles, which are **glomerular capillaries** surrounded by **Bowman's capsules.** The corpuscles are connected to **nephrons,** which are the tubules that are considered the functional units of the kidneys. The outer stripe of the outer medulla contains the thick ascending loops of Henle and collecting ducts, whereas the inner stripe contains the pars recta, thick and thin ascending loops of Henle and collecting ducts (Fig. 16.2). These empty urine into the calyces, and ultimately, the ureter, which leads to the bladder. Thus, a portion of the plasma fraction of blood entering the kidney is filtered through the **glomerular capillary membrane into Bowman's space,** flows into the nephrons, and becomes tubular fluid. After the tubular fluid is processed in the nephron, the remaining fluid (urine) flowing through the collecting ducts exits the renal pyramids into the minor calyces. The minor calyces combine to form the major calyces, which empty into the ureter (see Fig. 16.1*B*). The ureters lead to the bladder, where the urine is stored until excretion (micturition).

The Nephron

Each kidney contains more than 1 million nephrons. There are two populations of nephrons, **cortical** (or superficial) and **juxtamedullary** (deep) nephrons. Most of the nephrons are cortical (~80%), while ~20% are juxtamedullary. The populations are similar in that they are composed of the same structures, but differ in their location within the kidney and in the length of segments. The cortical nephrons originate from glomeruli in the upper and middle regions of the cortex, and their loops of Henle are short, extending only to the inner stripe of the outer medulla (see Fig. 16.2). The glomeruli of juxtamedullary nephrons are located deeper in the cortex (by

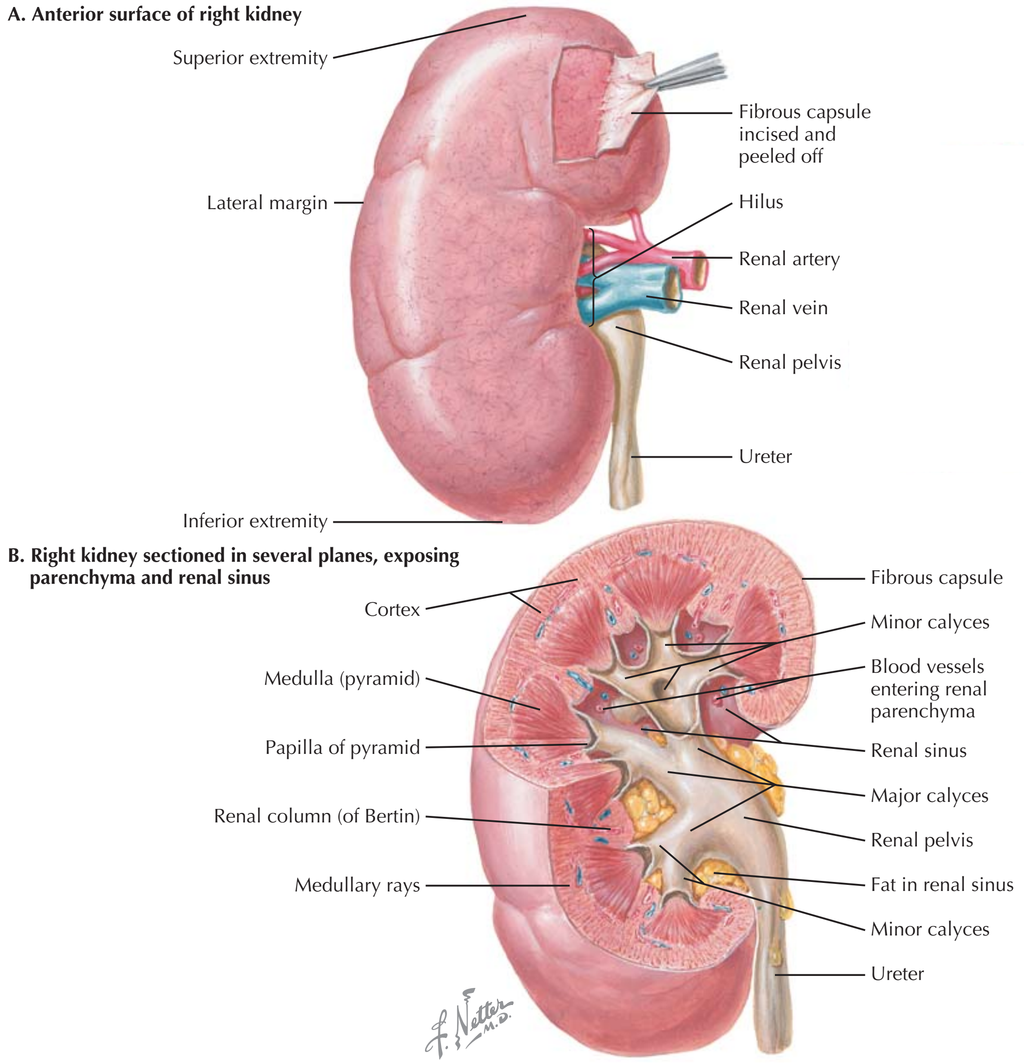


Figure 16.1 **Anatomy of the Kidney** The kidneys are bilateral organs with arterial blood supply from the abdominal aorta through the renal arteries *(A)*. The plasma is filtered at the glomeruli, which are located in the cortex. About 20% of the cardiac output enters the kidney (~1 L/min), and excess fluid and solutes are excreted as urine. The urine collects in the renal sinuses and exits the kidney via the ureter *(B),* which leads to the bladder for storage until elimination.

the medullary junction) and have long loops of Henle, extending deep into the inner medulla, forming the papillae.

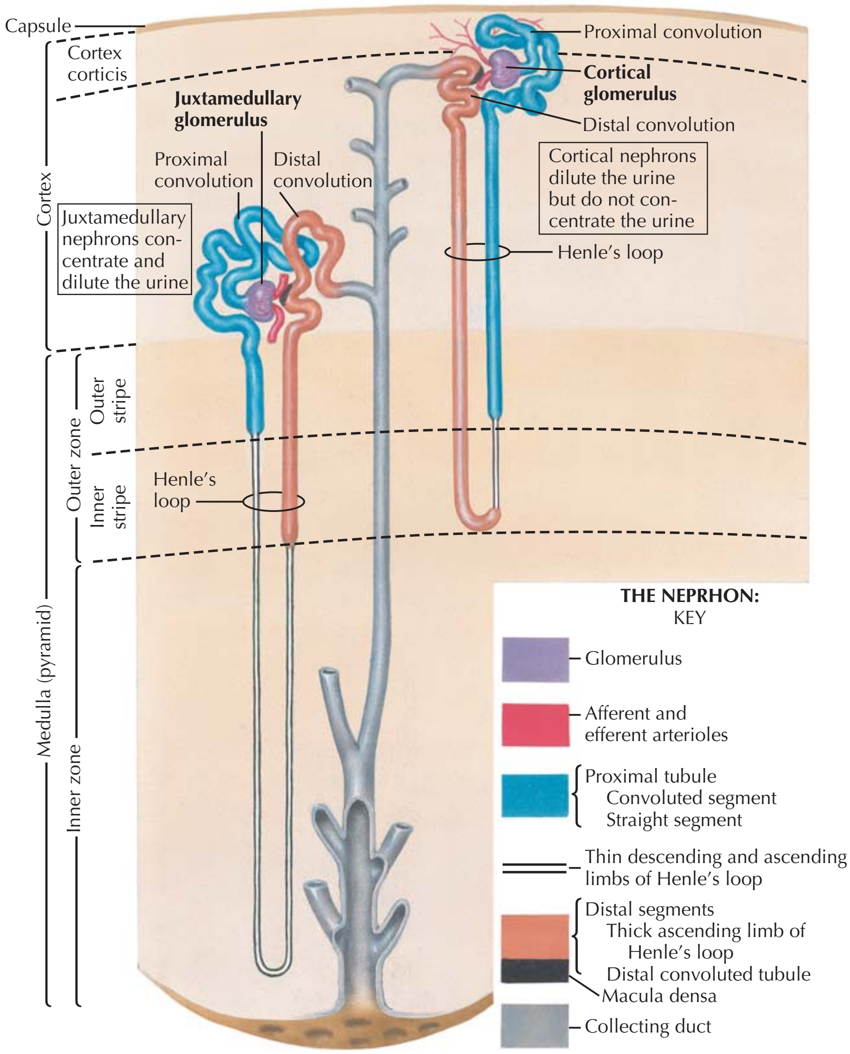
As stated, all nephrons have the same basic structures, but the location of the nephrons and the length of specific segments vary, with important consequences. The primary nephron segments are listed in sequential order in Table 16.1 with functions and distinctive characteristics.

Blood Flow

Blood flow to the kidneys (**renal blood flow, RBF**) is about 1 liter per minute (L/min), or ~20% of the cardiac output. The blood enters the kidneys via the renal arteries and follows the path shown:

→ interlobar arteries
→ arcuate arteries (at corticomedullary junction)
→ interlobar/cortical radial arteries
→ afferent arteriole (site of regulation)
→ **glomerular capillaries**
→ efferent arteriole
→ cortical peritubular capillaries (or vasa recta in deep nephrons)
→ venule
→ veins

The plasma fraction of the blood is filtered at the **glomerulus.** Blood enters the capillary from the *afferent arteriole* and exits the capillary by the *efferent arteriole*. Efferent arterioles associated with cortical nephrons, lead to the **peritubular capillar-**



Renal blood flow	1–1.2 L/min	Number of nerphons	2.5 million
Glomerular filtration rate	100–125 mL/min	Cortical	2.1 million
	140–180 L/day	Juxtamedullary	0.4 million
Urine flow rate	0.5–18 L/day		

Figure 16.2 Nephron Structure The nephron is the functional unit of the kidney, and the structure differs depending on the location of the glomerulus. The glomeruli of cortical (superficial) nephrons are located in the upper cortical zone of the kidney and have loops of Henle that extend only to the outer zone of the medulla. The glomeruli of the juxtamedullary (deep) nephrons are located at the cortico-medullary junction and have loops of Henle that extend deep into the inner medulla. There are ~5 times more cortical than juxtamedullary nephrons in the human kidney.

ies, which collect material reabsorbed from the nephrons; efferent arterioles of the juxtaglomerular nephrons lead to the **vasa recta** (straight vessels), which collect material reabsorbed from medullary tubules.

The Glomerulus

The glomerulus is a capillary system, from which an ultrafiltrate of plasma enters into Bowman's space (Fig. 16.3). The glomerular capillary has a fenestrated endothelium and **basement membrane,** which prevent filtration of blood cells, proteins, and most macromolecules into the glomerular ultrafiltrate. The glomerulus is surrounded by epithelial cells (**podocytes**) a single layer thick, which contribute to the filtration barrier. Filtration by the glomerulus occurs according to size and charge—because the basement membrane and podocytes are negatively charged, most proteins (also negatively charged) cannot be filtered. There are also **mesangial cells** that

Table 16.1 Nephron Segments: General Functions and Differences between Segments in Cortical vs. Juxtamedullary Nephrons

Segments	Description and General Functions of Segment	Characteristics in Cortical Nephrons	Characteristics in Juxtamedullary Nephrons
Glomerulus	The capillary net that filters plasma, making ultrafiltrate. Upon entering the proximal tubule, ultrafiltrate is called tubular fluid.	Located superficially, in the outer and mid-cortex; their efferent arterioles give rise to the peritubular capillaries.	Located deep in the cortex, by the medullary junction; efferent arterioles give rise to the vasa recta, which are adjacent to deep nephrons and aid in concentration of urine.
Proximal Convoluted Tubule	Has brush border villus membrane and is main site of reabsorption of solutes and water.	Shorter than proximal convoluted tubules in juxtamedullary nephrons.	Longer than in cortical nephrons, allowing relatively more reabsorption of solutes.
Proximal Straight Tubule	Additional reabsorption.	Much longer than in deep nephrons.	Shorter than in cortical nephrons.
Thin descending loop of Henle (tDLH)	Impermeable to solutes but permeable to water; thus, it *concentrates* tubular fluid as water diffuses out.	Much shorter than in deep nephrons.	Very long, forming pyramids, crucial for concentrating tubular fluid.
Thick ascending loop of Henle (TALH)	Impermeable to water, but has Na^+-K^+-$2Cl^-$ transporters that reabsorb more solutes and *dilute* the tubular fluid. Sets up and maintains interstitial concentration gradient.	Longer than deep nephrons, dilutes tubular fluid.	Dilutes tubular fluid and is critical in producing the large concentration gradient in the inner medulla.
Distal convoluted tubule	Electrolyte modifications; aldosterone acts on late distal segments.	Similar in cortical and deep nephrons.	Similar in cortical and deep nephrons.
Collecting ducts (CD)	Site of free water reabsorption through water channels (aquaporins) controlled by ADH. CDs are also important for acid–base balance: the **α-intercalated cells** allow H^+ secretion; **β-intercalated cells** have HCO_3^-/Cl^- exchangers, which allow HCO_3^- secretion, when necessary.	The cortical collecting ducts (CCD) reabsorb some Na^+ and Cl^- and secrete K^+ (from aldosterone-sensitive principal cells). Less effect on urine concentration compared with deep nephrons because the ducts do not extend far into medulla. CCDs also have **α**- and **β**-intercalated cells for acid–base regulation.	Because they extend deep into the medulla, the final concentration of urine occurs here. The inner medullary collecting ducts (IMCD) have principal cells (with aldosterone-sensitive Na^+ and K^+ channels), as well as intercalated cells (as seen in CCDs). Medullary CDs are a key site of ADH-dependent urea reabsorption, which contributes to the high medullary interstitial fluid osmolarity.

ADH, antidiuretic hormone.

support the glomerulus but can also contract, decreasing surface area for filtration.

The Juxtaglomerular Apparatus

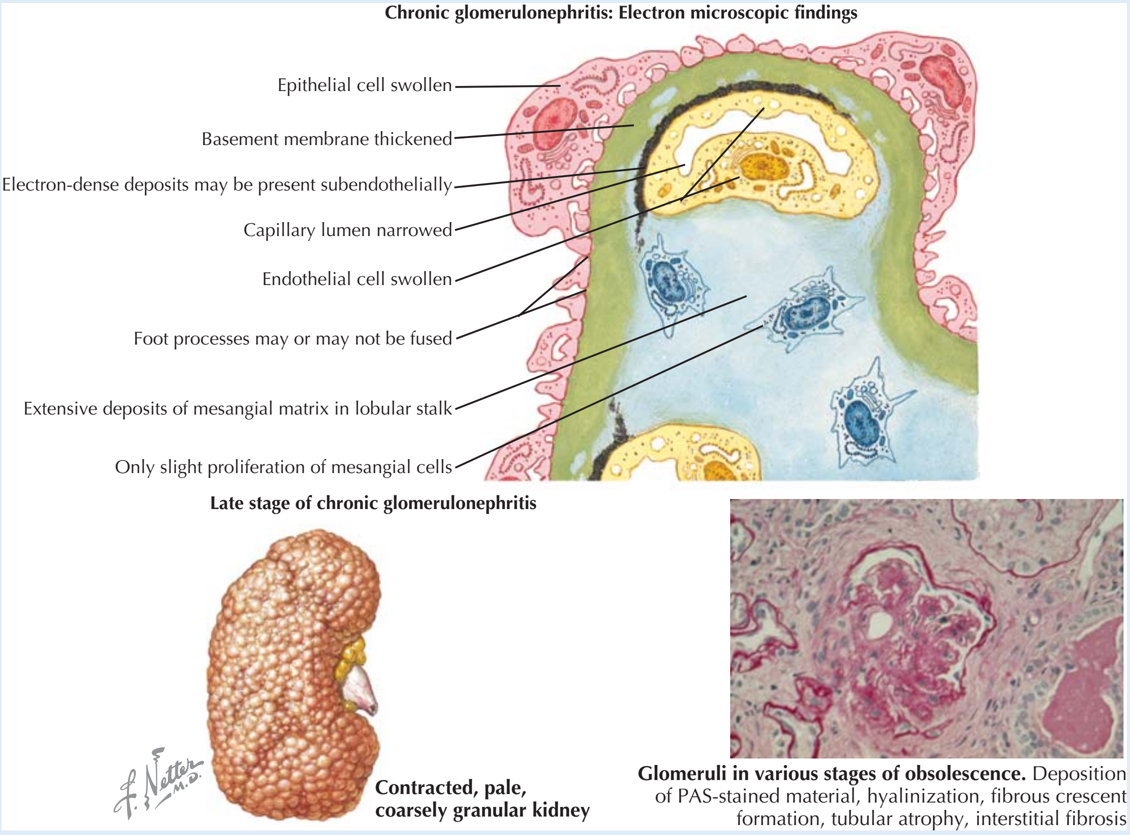
Another important structural and functional aspect is the juxtaglomerular apparatus—this is the area where the distal convoluted tubule returns to its "parent" glomerulus. At this site, specialized **macula densa** cells are in contact with the distal convoluted tubule and afferent arteriole, forming the **juxtaglomerular apparatus** (see Fig. 16.3). The macula densa cells of the juxtaglomerular apparatus are important in sensing tubular fluid flow and sodium delivery to the distal nephron, and because of their proximity to the afferent arteriole, macula densa cells can regulate renal plasma flow and **glomerular filtration rate** (**GFR**) (autoregulation). Macula densa cells also participate in the regulation of the release of the enzyme **renin** from juxtaglomerular cells adjacent to the afferent arterioles. The renin secretion aids in fluid and electrolyte homeostasis (see Chapter 19). Macula densa cells also receive input from adrenergic nerves through β_1-receptors.

CLINICAL CORRELATE
Glomerulonephritis

The glomerulus is a key site for renal damage. Diseases and drugs that damage the glomerular basement membrane reduce the negative charge and allow large proteins (especially albumin) to be filtered. Because there is no mechanism for reabsorbing large proteins in the nephron, the protein is excreted in the urine (proteinuria). In addition, diseases (such as diabetes) that increase mesangial matrix deposition increase rigidity and decrease area of filtration of the glomerulus, reducing renal function.

Acute glomerulonephritis is usually caused by different factors in children and adults. In children, a common cause is streptococcal infection. In adults, acute glomerulonephritis can arise as a complication from drug reactions, pneumonia, immune disorders, and mumps. Acute glomerulonephritis can be asymptomatic (about 50% of cases) or can be associated with edema, low urine volume, headaches, nausea, and joint pain. Treatment is aimed at reducing the inflammation, usually with steroids or immunosuppressive drugs, while determining and addressing the cause, when possible. In most cases patients recover completely.

In contrast, **chronic glomerulonephritis** is associated with long-term inflammation of glomerular capillaries, resulting in thickened basement membranes, swollen epithelial cells, and narrowing of the capillary lumen. Major causes of chronic glomerulonephritis are diabetes, lupus nephritis, focal segmental glomerulosclerosis, and IgA nephropathy. The rate of progression of kidney damage to chronic renal failure (GFR less than 10 to 15 mL/min) is widely variable and can take as few as 5 years or more than 30 years, depending on the overall cause of the inflammatory process. Chronic glomerulonephritis can lead to other major systemic complications including hypertension, heart failure, uremia, and anemia. Treatment is dependent on the cause of the damage, and in the case of diabetes-induced disease, angiotensin II receptor blockers or angiotensin-converting enzyme inhibitors are beneficial in slowing the renal damage. As the damage progresses toward end-stage renal failure, the GFR is insufficient to rid the body of waste, and uremia is one of the results. Patients usually start hemodialysis when their GFR is less than 20 mL/min. Dialysis can be used for years, although many patients opt for renal transplantation, which is a common procedure.



Chronic Glomerulonephritis The upper panel illustrates key features of chronic glomerular damage, including swollen epithelial cells, a grossly thickened basement membrane, fused foot processes, and increased matrix proteins. These abnormalities destroy the normal filtration barriers. The lower left panel depicts the effects of severe glomerulonephritis on the whole kidney, and the lower right panel gives a representative micrograph of damaged glomeruli.

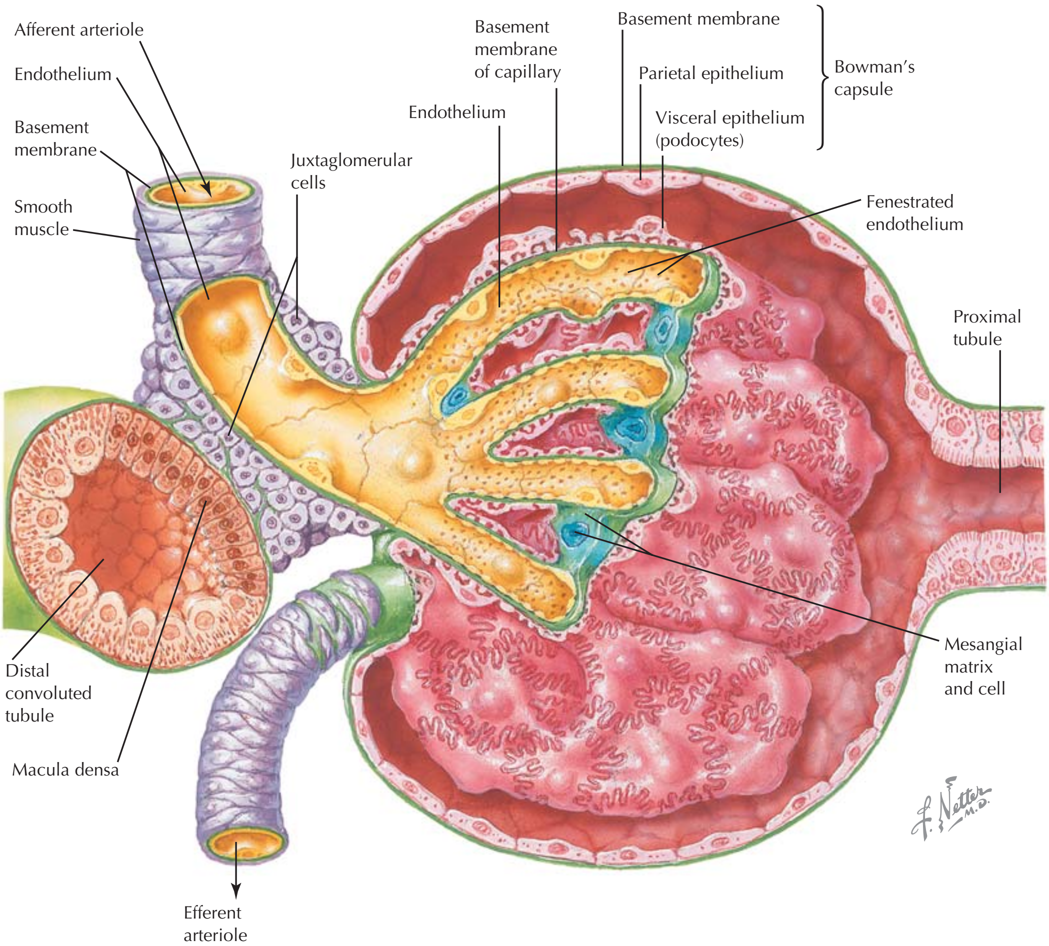


Figure 16.3 **Anatomy of the Glomerulus** Plasma is filtered at the glomerular capillaries into Bowman's space, and the ultrafiltrate then flows into the proximal tubule. The glomerular endothelial barrier prevents filtration of the cellular elements of the blood, so the ultrafiltrate does not contain blood cells or plasma proteins. The cells of the macula densa are in contact with the afferent arteriole through the juxtaglomerular cells, forming the juxtaglomerular apparatus. The macula densa monitors NaCl delivery to the distal tubule and regulates renal plasma flow (autoregulation).

Renal Plasma Flow

While whole blood enters the renal arteries, only *plasma* is filtered at the glomerular capillaries, and thus, when discussing glomerular filtration, **renal plasma flow** (**RPF**) is an important factor. RPF can be determined by the following equation:

$$RPF = RBF \times (1 - HCT)$$

In the normal adult male, RBF = ~1 L/min, and hematocrit (HCT) is ~40% (0.4). Thus,

$$RPF = 1\ L/min \times 0.6 = 600\ mL/min$$

To determine the **effective renal plasma flow** (**EPRF**), which is the plasma flow entering the glomeruli and available for filtration, the plasma clearance of the inorganic acid **para-aminohippurate** (**PAH**) is used. PAH is filtered at the glomeruli, and under normal circumstances the remaining PAH in the peritubular capillaries is *secreted* into the proximal tubule, so that essentially no PAH enters the renal vein (Fig. 16.4 and see "Analysis of Renal Function" Clinical Correlate).

GLOMERULAR FILTRATION: PHYSICAL FACTORS AND STARLING FORCES

Glomerular filtration is determined by the Starling forces and the permeability of the glomerular capillaries to the solutes in the plasma. In general, with the exception of formed elements (red blood cells, white blood cells, platelets) and most proteins, plasma is available for filtration at the glomerular capillaries. Because the molecules must travel through several barriers to move from the capillary lumen to Bowman's space

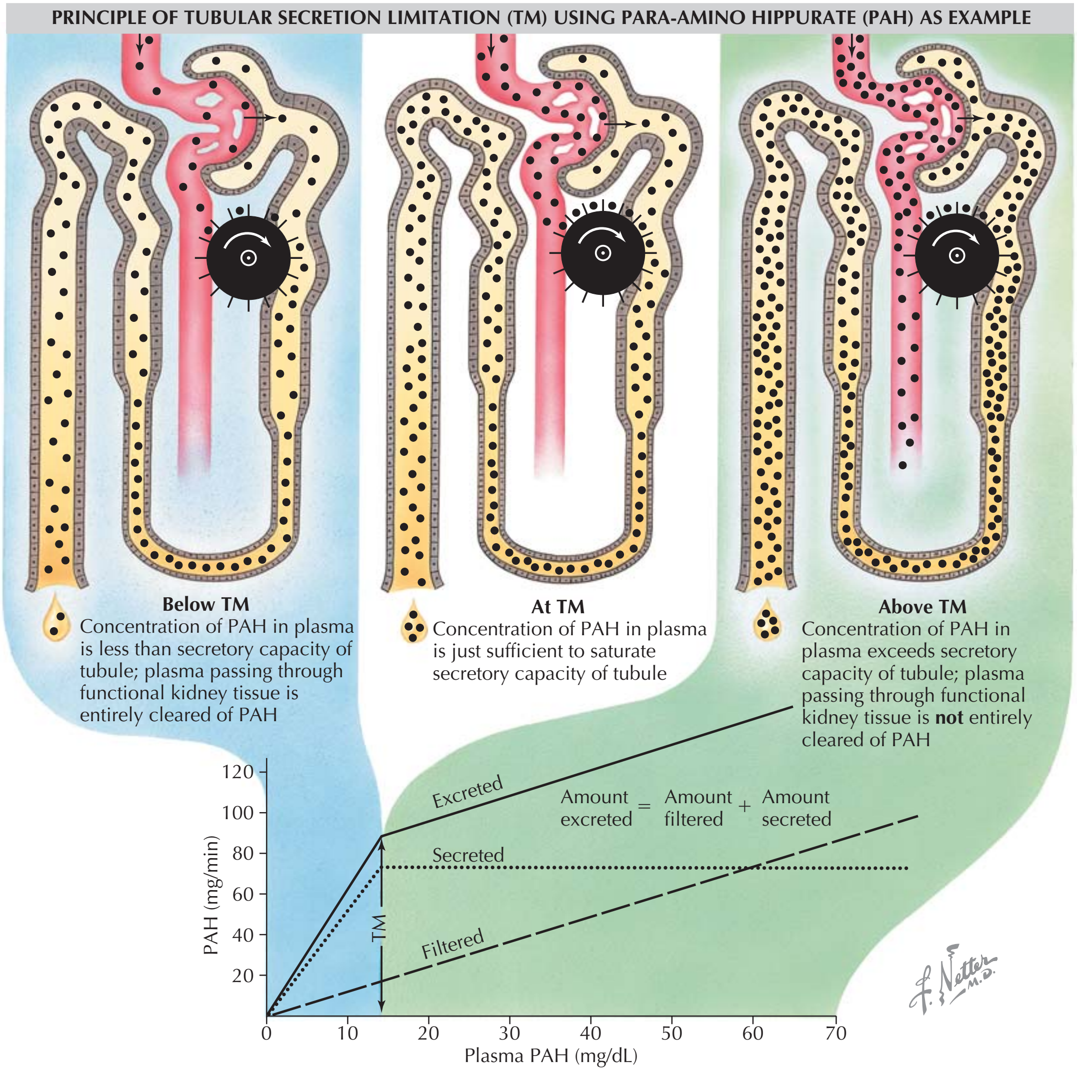


Figure 16.4 Renal Handling of Para-amino Hippurate (PAH) PAH is filtered at the glomerulus and also secreted into the proximal tubule. When the plasma concentration of PAH is below the tubular transport maximum (TM), PAH is effectively cleared from the blood entering the kidney. However, if the plasma concentration exceeds the TM, PAH is not entirely removed and is found in the renal vein.

(fenestrated epithelium → basement membrane → between podocytes → filtration slit → Bowman's space), there are size limitations, and ultimately the effective pore size is ~30 Å. Small molecules such as water, glucose, sucrose, creatinine, and urea are freely filtered. As molecular size increases, or net negative charge of molecules increases (for example, among proteins), filtration becomes increasingly restricted.

Starling forces govern fluid movement into or out of the capillaries (see Chapter 1). The pressures that determine glomerular filtration dynamics are glomerular capillary hydrostatic pressure (HP_{GC}) forcing fluid out of the capillary, glomerular capillary oncotic pressure (π_{GC}) attracting fluid into the glomerular capillary, Bowman's space hydrostatic pressure (HP_{BS}) opposing capillary hydrostatic pressure, and Bowman's space oncotic pressure (π_{BS}) attracting fluid into Bowman's space (typically there is negligible protein in the

Myoglobin, a small protein that is released from muscle following damage, is only 20 Å, but its shape restricts free passage, and only about 75% is filtered. Most proteins are negatively charged or of high molecular weight and will not be filtered unless there is damage to the glomerular barriers, or the negative charge of the protein is affected by viral or bacterial processes. In those cases, protein will enter the renal tubule and be excreted in urine (proteinuria).

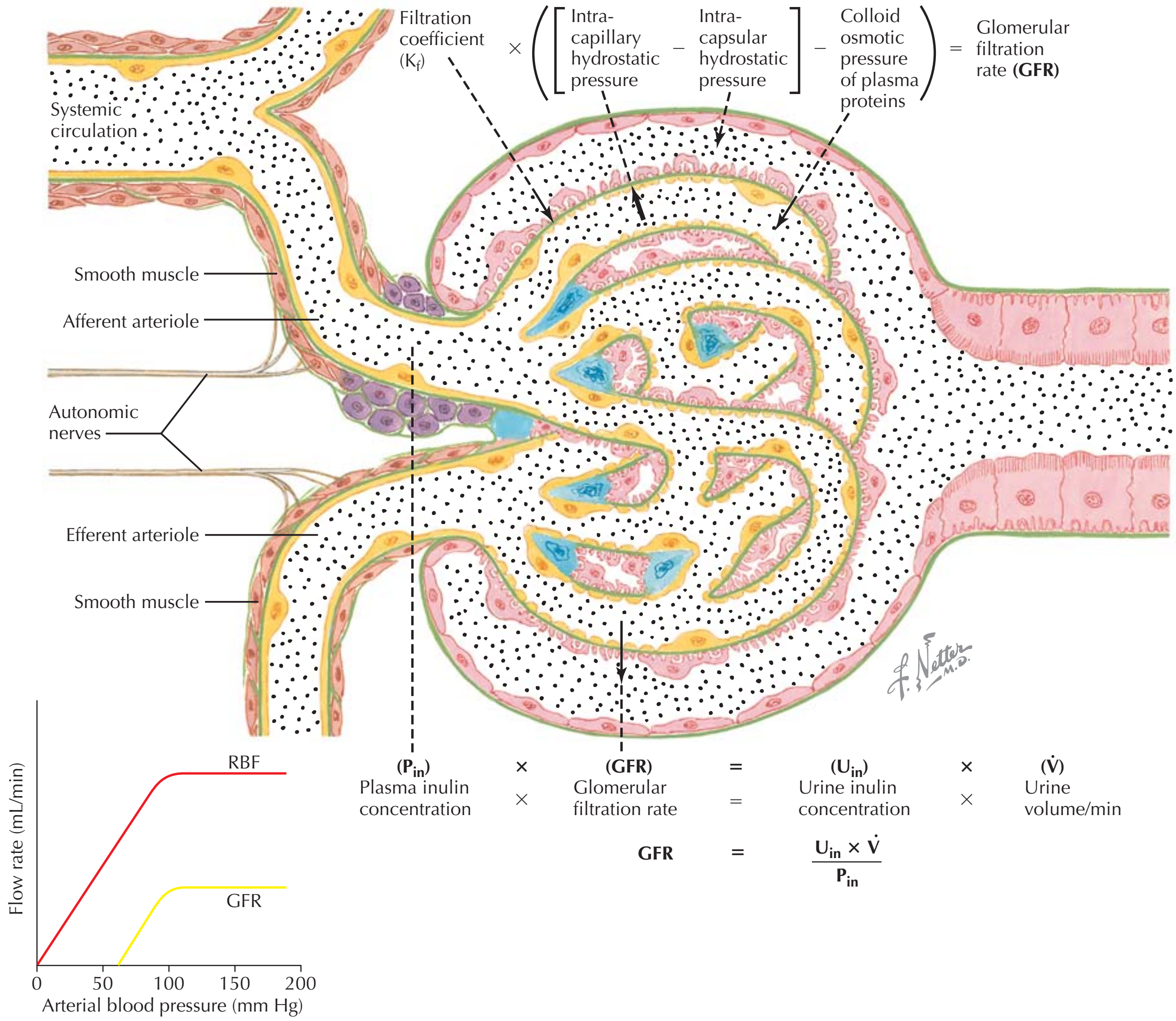


Figure 16.5 **Glomerular Filtration** Blood enters the glomerular capillaries from the afferent arterioles, and ~20% of the fluid is filtered into the nephrons (filtration fraction). The glomerular filtration rate (GFR) can be described on the basis of the forces governing filtration *(upper equation),* or from the clearance of inulin *(lower equation).* The graph illustrates that renal blood flow (RBF) and GFR remain fairly constant over a wide range of mean arterial blood pressures (MAP)—this occurs in part through autoregulation and tubulo-glomerular feedback.

Bowman's space, so π_{BS} is not significant). Thus, assuming π_{BS} is zero,

$$\text{Net filtration pressure} = (HP_{GC} - HP_{BS}) - \pi_{GC}$$

The glomerular capillaries are different from other capillaries (which have significantly reduced pressures at the distal end of the capillary), because the efferent arteriole (at the other end of the glomerulus) can constrict and maintain pressure in the glomerular capillary. Thus, there is very little reduction in HP_{GC} through the capillary, and filtration can be maintained along its entire length. Afferent and efferent arteriolar resistance can be controlled by neural influences, circulating hormones (angiotensin II), myogenic regulation, and tubuloglomerular feedback signals, allowing control of glomerular filtration by both intrarenal and extrarenal mechanisms.

Glomerular Filtration Rate

Glomerular filtration rate (GFR) is considered the benchmark of renal function. GFR is the amount of plasma (without protein and cells) that is filtered across all of the glomeruli in the kidneys, per unit time. In a normal adult, GFR is ~100 to 125 mL/min, with men having higher GFR than women. Many factors contribute to the regulation of GFR, which can be maintained at a fairly constant rate, despite fluctuations in mean arterial blood pressure (MAP) from 80 to 180 mm Hg (Fig. 16.5).

GFR is determined by the net filtration pressure, as well as physical factors associated with the glomeruli, or *Kf* (hydraulic permeability and total surface area, which reflects nephron number and size). The equation is:

$$GFR = Kf\,[(HP_{GC} - HP_{BS}) - \pi_{GC}]$$

Maintaining normal GFR is critical for eliminating excess fluid and electrolytes from the blood and maintaining overall homeostasis. Significant alteration of any of the parameters in the equation above can affect GFR. For example, a hemorrhage that reduces MAP below 80 mm Hg may decrease HP_{GC} enough to dramatically decrease or stop filtration. Filtration can also be reduced if the HP_{BS} is increased (for example, during distal blockage by kidney stones), or if *Kf* is reduced (for example, in glomerulosclerosis).

In general, the nephrons are associated with filtration, reabsorption, secretion, and excretion:

- The **filtered load** (FLx) of a substance (the amount of a specific substance filtered per unit time) is equal to the plasma concentration of the substance (Px) times GFR:

$$FLx = Px \times GFR$$

- The **urinary excretion** (Ex) of a substance is the urine concentration of the substance (Ux) times the volume of urine produced per unit time ($\dot{V}$):

$$Ex = Ux \times \dot{V}$$

- Most substances are reabsorbed (to some extent); reabsorption rate of a substance (Rx) is equal to the filtered load (FLx) of the substance minus the urinary excretion of a substance (Ex):

$$Rx = FLx - Ex$$

- Select substances are actively secreted (e.g., creatinine, PAH, H^+, K^+). The secretion rate of a substance (Sx) is equivalent to the excretion rate minus the filtered load of the substance:

$$Sx = Ex - FLx$$

Renal handling of key substances will be discussed in Chapter 17.

RENAL CLEARANCE

Because GFR is a primary measure of the health of kidney function, GFR is routinely analyzed. This can be done in several ways. The physical factors and pressures can all be measured experimentally, but this is not practical in humans. Instead, the principle of **renal clearance** is utilized. Renal clearance is the *volume of plasma cleared of a substance per unit time*. The clearance equation incorporates the urine and plasma concentrations of the substance, and the urine flow rate and is usually reported in mL/min or L/day:

$$Cx = (Ux \times \dot{V})/Px$$

This equation can be used to easily determine the GFR: the clearance of a substance is equated to the GFR if the substance is freely filtered, but not reabsorbed or secreted. In this case, *the amount filtered will equal the amount excreted* [FLx = Ex], and thus:

$$\text{since } FLx = Px \times GFR \text{ and } Ex = Ux \times \dot{V}$$

$$\text{when } \mathbf{Flx = Ex}$$

then,

$$Px \times GFR = Ux \times \dot{V}$$

and, rearranging the equation,

$$GFR = (Ux \times \dot{V})/Px$$

Thus, for such a substance, GFR = Cx.

Although there is no endogenous substance that exactly meets these requirements (i.e., the substance is freely filtered, but not reabsorbed or secreted, and, therefore, FLx = Ex), the polyfructose molecule **inulin** does meet these criteria. It is not broken down in the blood, is freely filtered, and is not reabsorbed or secreted by the kidney. To measure inulin clearance (and thereby determine GFR), inulin is infused intravenously, and when a stable plasma level is achieved, timed urine

The RPF feeds the glomerular capillaries, but not all of the plasma presented to the capillaries is filtered. The filtration fraction (FF) is the proportion of the RPF that becomes glomerular filtrate:

$$FF = GFR/RPF$$

In the normal adult, FF = (125 mL/min ÷ 600 mL/min) × 100 = ~20%, so ~20% of the plasma entering the kidneys is filtered. At the individual nephrons, the unfiltered plasma exits the efferent arteriole to the peritubular capillaries.

If the clearance of inulin (C_{in}) is 100 mL/min, it means that 100 mL of plasma is completely cleared of inulin each minute. Contrast that to the clearance of glucose, which is 0 mL/min in a normal person, indicating that no plasma is cleared of glucose (and therefore there is no glucose in the urine). The renal clearance of any filtered substance can be calculated, and when the clearance is compared with the GFR, it gives a general idea of whether there was net reabsorption or net secretion of the substance—this is because the GFR is the total rate of filtration that is occurring at any given time.

- If the clearance of X is less than the GFR, there is net reabsorption.
- If the clearance of X is greater than the GFR, there is net secretion, because more was cleared from the plasma than can be accounted for by GFR alone.

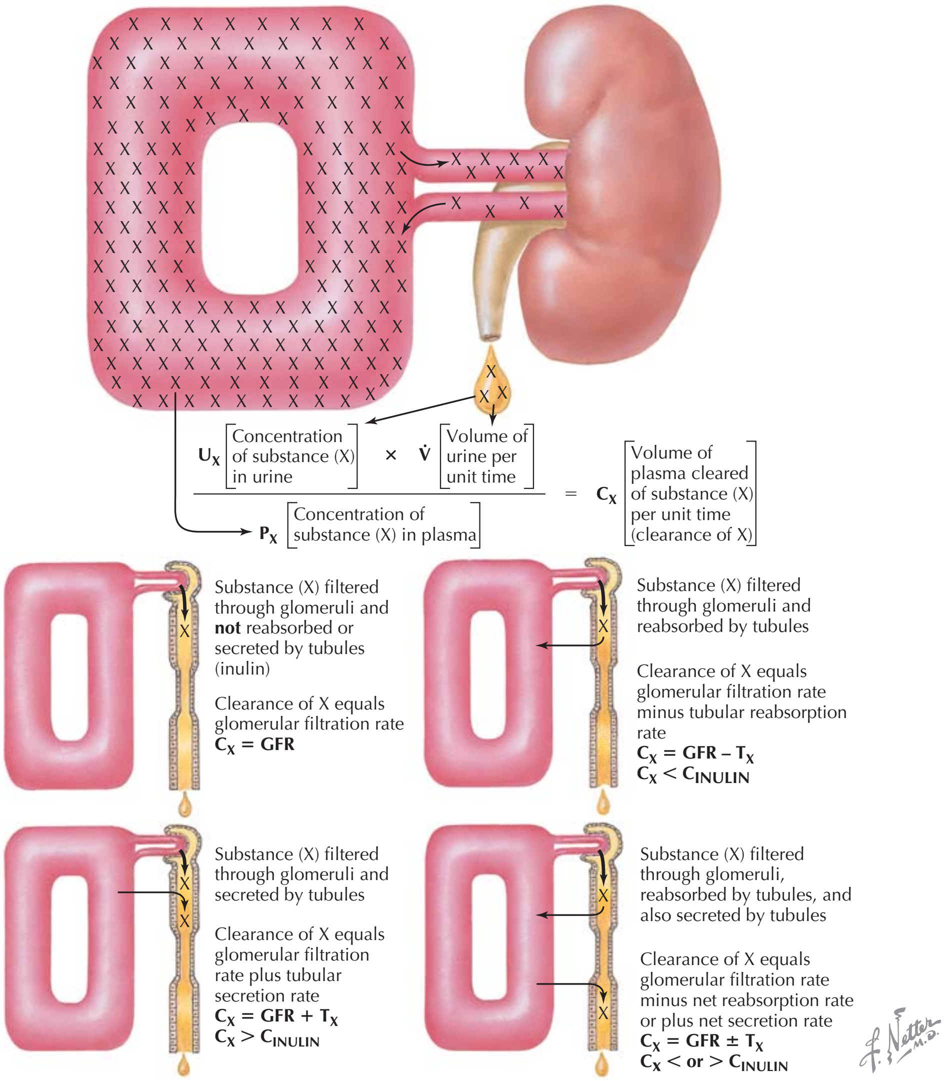


Figure 16.6 **Renal Clearance Principle** "Clearance" describes the volume of plasma that is cleared of a substance per unit time. The renal clearance of a substance provides information on how the kidney handles that substance. Since inulin is freely filtered, and not reabsorbed or secreted, all of the filtered inulin is excreted in the urine. Thus, C_{in} is equated with the glomerular filtration rate (GFR), and the net handling of other substances can be determined, depending on whether their clearance is greater than (net secretion), less than (net reabsorption), or equal to C_{in}.

Plasma creatinine is used clinically to estimate GFR. In most cases, the body produces creatinine at a constant rate, so the excretion rate is also constant. Since GFR is equated with the clearance of creatinine [GFR = $(U_{Cr} \times \dot{V}) \div P_{Cr}$], if creatinine excretion ($U_{Cr} \times \dot{V}$) is constant, the GFR is proportional to $1/P_{Cr}$. So, when the GFR decreases, less creatinine is filtered and excreted, and plasma creatinine builds up. As a clinical application, this allows a rapid approximation of the GFR by simply analyzing the P_{Cr}. P_{Cr} is normally ~1 mg%, so GFR is proportional to 1/1, or 100%. If P_{Cr} rises to 2, GFR is ½, or 50%, and so on.

collections are made. The calculated clearance of inulin can be equated to the GFR (see Fig. 16.5):

$$C_{inulin} = GFR$$

Infusing inulin to determine clearance is not routinely used because of the invasive nature of the procedure. Instead, the renal clearance of the *endogenous* substance **creatinine** is used to approximate GFR. Creatinine is a by-product of muscle metabolism and is freely filtered by the kidneys. It is not reabsorbed, but there is ~10% secretion into the renal tubules from the peritubular capillaries, and thus, creatinine clearance overestimates GFR by ~10% (Fig. 16.6).

REGULATION OF RENAL HEMODYNAMICS

Regulation of the GFR occurs by changes in blood flow to the glomeruli via intrinsic feedback systems, hormones, vasoactive substances, and renal sympathetic nerves.

Intrinsic systems include the myogenic mechanism and tubuloglomerular feedback (TGF). Utilizing the **myogenic mechanism,** the renal arteries and arterioles respond directly to increases in systemic blood pressure by constricting, thereby maintaining constant filtration pressure in the glomerular capillaries. **Tubuloglomerular feedback (TGF)** is a regulatory mechanism that involves the macula densa of the juxtaglomerular apparatus. The kidney is unique in that the glomerular capillaries have arterioles (resistance vessels) on *either end* of the capillary network. Constriction of the afferent or efferent arterioles can elicit immediate effects on the HP_{GC}, controlling GFR. Because the juxtaglomerular apparatus functionally associates the distal tubule to the afferent arteriole, the tubular flow past the macula densa can control afferent arteriolar resistance (see Fig. 16.3). Decreases in flow and tubular fluid sodium concentration in the distal tubule will decrease afferent arteriolar resistance and increase GFR in that nephron; conversely, if distal tubular flow or osmolarity is high, TGF will increase afferent arteriolar resistance, decreasing GFR. These systems allow minute-to-minute regulation of GFR over a wide range of systemic blood pressures (MAP 80 to 180 mm Hg).

Many substances (including nitric oxide and endothelin) regulate renal hemodynamics, but this section will focus on the renin-angiotensin-aldosterone system (RAAS), atrial natriuretic peptide (ANP), sympathetic nerves/catecholamines, and intrarenal prostaglandins. Although angiotensin II and the sympathetic nervous system are activated to preserve systemic blood pressure, the kidneys will respond to excessive constriction by intrarenal autoregulation, preserving blood flow to the glomeruli. This balance between extrarenal and intrarenal control is necessary to maintain proper GFR.

Control of renal hemodynamics occurs through the following neurohumoral and paracrine mechanisms:

- The **renin-angiotensin-aldosterone system (RAAS)** is activated in response to low renal vascular flow. Renal vascular baroreceptors stimulate **renin** secretion by the juxtaglomerular cells at the ends of the afferent arterioles. This, in addition to the modulation of renin secretion by the macula densa, will activate the RAAS (see detailed description in Chapter 19). The renin will act locally and through the systemic circulation to produce angiotensin II, and thus control GFR.
- **Angiotensin II** exerts both direct and indirect effects on the GFR. It is a vasoconstrictor, and in the kidneys, it acts directly on the renal arteries, and to a greater extent at the afferent and efferent *arterioles,* increasing resistance, reducing HP_{GC}, and decreasing GFR; angiotensin II actually has greater effect on the efferent arteriole than afferent arteriole. At the same time, it can constrict *glomerular* mesangial cells, reducing Kf, and thus, GFR.
- **Atrial natriuretic peptide (ANP)** is released from the right cardiac atrial myocytes in response to stretch (at high blood volume). To regulate GFR, ANP dilates the afferent arteriole, and constricts the efferent arteriole, increasing HP_{gc}, and thus, GFR. The enhanced flow increases sodium and water excretion, reducing blood volume.
- **Sympathetic nerves** and **catecholamine secretion** (NE and Epi) are stimulated in response to reductions in systemic blood pressure and cause vasoconstriction of the renal arteries and arterioles. At tonic levels of sympathetic nerve activity, the intrarenal systems will counteract this effect, to ensure the kidney vasculature remains dilated, preserving GFR. During high sympathetic nerve activity (hemorrhage, strenuous exercise), sympathetic nerve activity overrides the intrarenal regulatory mechanisms and reduces renal blood flow and GFR.
- **Intrarenal prostaglandins** (PGE_2 and prostacyclin [PGI_2]) are vasodilators and serve to counteract primarily angiotensin II–mediated vasoconstriction, acting at the level of the arterioles and glomerular mesangial cells. Nonsteroidal anti-inflammatory drugs (NSAIDs) such as aspirin will block prostaglandin synthesis and restrict the compensatory renal vasodilation.

With blood loss from hemorrhage, the sympathetic nervous system (SNS) and hormone systems (RAAS, antidiuretic hormone [ADH], aldosterone) are activated to preserve systemic blood pressure, and prevent fluid loss. If MAP falls below 80 mm Hg, the high level of vasoconstriction will overwhelm the intrarenal regulation of GFR, and GFR will drop. This can result in acute renal failure (GFR < 25 mL/min) if blood volume is not restored quickly.

CLINICAL CORRELATE
Analysis of Renal Function

This correlate will focus on the variety of calculations associated with renal function and give examples of their solution.

A constant amount of inulin (in isotonic saline) was infused intravenously to a healthy 25-year-old male. After 3 hours, the man emptied his bladder completely, and then urine was collected after another 2 hours. A blood sample was obtained at the time of urine collection. Blood and urine were analyzed, with results shown below. Analyses of several parameters of renal function were performed.

	Urine	Plasma
Inulin concentration	1000 mg%	20 mg%
Creatinine concentration	1375 mg%	25 mg%
PAH concentration	300 mg%	1 mg%
Sodium concentration	2.5 mEq/L	140 mEq/L

Urine volume (UV) = 240 mL
Urine collection time = 2 hours
Hematocrit (HCT) = 0.42

The following parameters can be calculated:

Urinary flow rate ($\dot{V}$)**,** the rate at which urine is produced. Urine flow is dependent on general fluid homeostasis and fluid intake. Under normal circumstances, if fluid intake is increased, urine flow will increase. If a person ingests ~3 L of fluid in food and drink, the urinary losses will be slightly less, with the balance made up by insensible losses (breathing, sweating).

$$\dot{V} = \text{urine volume/time} = 240\ \text{mL}/120\ \text{min} = 2\ \text{mL/min}$$

Glomerular filtration rate (**GFR**)**,** the volume of plasma filtered by the glomeruli per unit time. Normal GFR in an adult is ~100 mL/min, or ~144 L/day. The GFR in men is typically higher than in women.

GFR is determined by inulin clearance:

$$C_{in} = (U_{in} \times \dot{V})/P_{in} = (1000\ \text{mg\%} \times 2\ \text{mL/min})/20\ \text{mg\%} = 100\ \text{mL/min}$$

GFR can also be determined by creatinine clearance, which overestimates GFR by ~10% because of creatinine secretion:

$$C_{cr} = (U_{cr} \times \dot{V})/P_{cr} = (1375\ \text{mg\%} \times 2\ \text{mL/min})/25\ \text{mg\%} = 110\ \text{mL/min}$$

Effective renal plasma flow (**eRPF**)**,** the fraction of the renal plasma flow entering the glomeruli and available for filtration. *eRPF is equated with the clearance of PAH:*

$$\text{eRPF} = C_{PAH} = (300\ \text{mg\%} \times 2\ \text{mL/min})/1\ \text{mg\%} = 600\ \text{mL/min}$$

Effective renal blood flow (eRBF), the fraction of renal blood flow entering the glomeruli. It is usually ~20% of cardiac output.

$$\text{eRBF} = (\text{eRPF})/(1 - \text{HCT}) = 600\ \text{mL/min}/(1 - 0.42) = 1034\ \text{mL/min, or } 1.034\ \text{L/min}$$

Filtration fraction (**FF**)**,** the fraction of the renal plasma flow that is filtered per unit time.

$$\text{FF} = \text{GFR/RPF} = (100\ \text{mL/min})/(600\ \text{mL/min}) = 0.17\text{, or 17\% of the RPF entering the kidney was filtered per minute}$$

Filtered load of sodium (**FL_{Na}**)**,** the amount of plasma sodium that is filtered per unit time.

$$FL_{Na} = \text{Plasma Na} \times \text{GFR} = 140\ \text{mEq/L} \times 100\ \text{mL/min} = 14\ \text{mEq/min}$$

Urinary excretion of sodium (**$U\dot{V}_{Na}$ or E_{Na}**)

$$U\dot{V}_{Na} = \text{Urine concentration of Na} \times \dot{V} = 2.5\ \text{mEq/L} \times 2\ \text{mL/min} = 0.005\ \text{mEq/min}$$

Reabsorbed sodium (**R_{Na}**)

$$R_{Na} = FL_{Na} - U\dot{V}_{Na} = 14\ \text{mEq/min} - 0.005\ \text{mEq/min} = 13.095\ \text{mEq/min}$$

Fractional excretion of sodium (**FE_{Na}**)**,** the fraction of filtered sodium that is excreted. Usually 99+% of filtered sodium is reabsorbed, so less than 1% of the amount filtered is excreted.

$$FE_{Na} = [(U/P)_{Na}/(U/P)_{in}] \times 100 = [(2.5/140)/(1000/20)] \times 100 = 0.035\%$$

Fractional reabsorption of sodium (**FR_{Na}**)**,** the fraction of filtered sodium that is reabsorbed back into the capillaries.

$$FR_{Na} = [1 - (E_{Na}/FL_{Na})] \times 100 = [1 - (0.005/14)] \times 100 = 99.97\%$$

CHAPTER 17 Renal Transport Processes

GENERAL OVERVIEW OF RENAL TRANSPORT

When the plasma filtered into Bowman's space enters the proximal tubule, the process of reabsorption begins. In general, nephrons reabsorb the majority of the fluid and solutes that pass though them, with the proximal tubule having the greatest reabsorptive function, and the distal sites fine-tuning the process. In addition, there is secretion of select substances from the peritubular capillaries into different segments of the renal tubule.

The proximal tubule (PT) is the site of bulk reabsorption of fluid and nutrients. The proximal tubule is composed of three segments, S1, S2, and S3, which differ in the depth of the brush border and amount of mitochondria in the PT cells. This allows for a high capacity for reabsorption. From S1 to S3 segments, the brush border becomes progressively deeper and the high concentration of cellular mitochondria observed in the S1 segment decreases. The high number of mitochondria in the S1 is consistent with a high rate of active transport in that segment. As the filtrate is reabsorbed, and less is present in the tubule in subsequent segments, the deeper brush border increases surface area, which enhances continued reabsorption.

SODIUM-DRIVEN SOLUTE TRANSPORT

Sodium, Chloride, and Water

Sodium is the major extracellular cation, and regulation of its levels is necessary for maintenance of general fluid and electrolyte homeostasis (Chapter 1). As seen with the intestinal absorption of essential nutrients (see Section 6), sodium is also a major driving force for the renal reabsorption of fluid, electrolytes, and a variety of nutrients. As sodium transporters carry sodium and other solutes, they generate the driving force for water reabsorption. When the water leaves the tubule, the concentration of additional electrolytes and solutes in the tubular fluid increases, providing gradients for their diffusion into the cell.

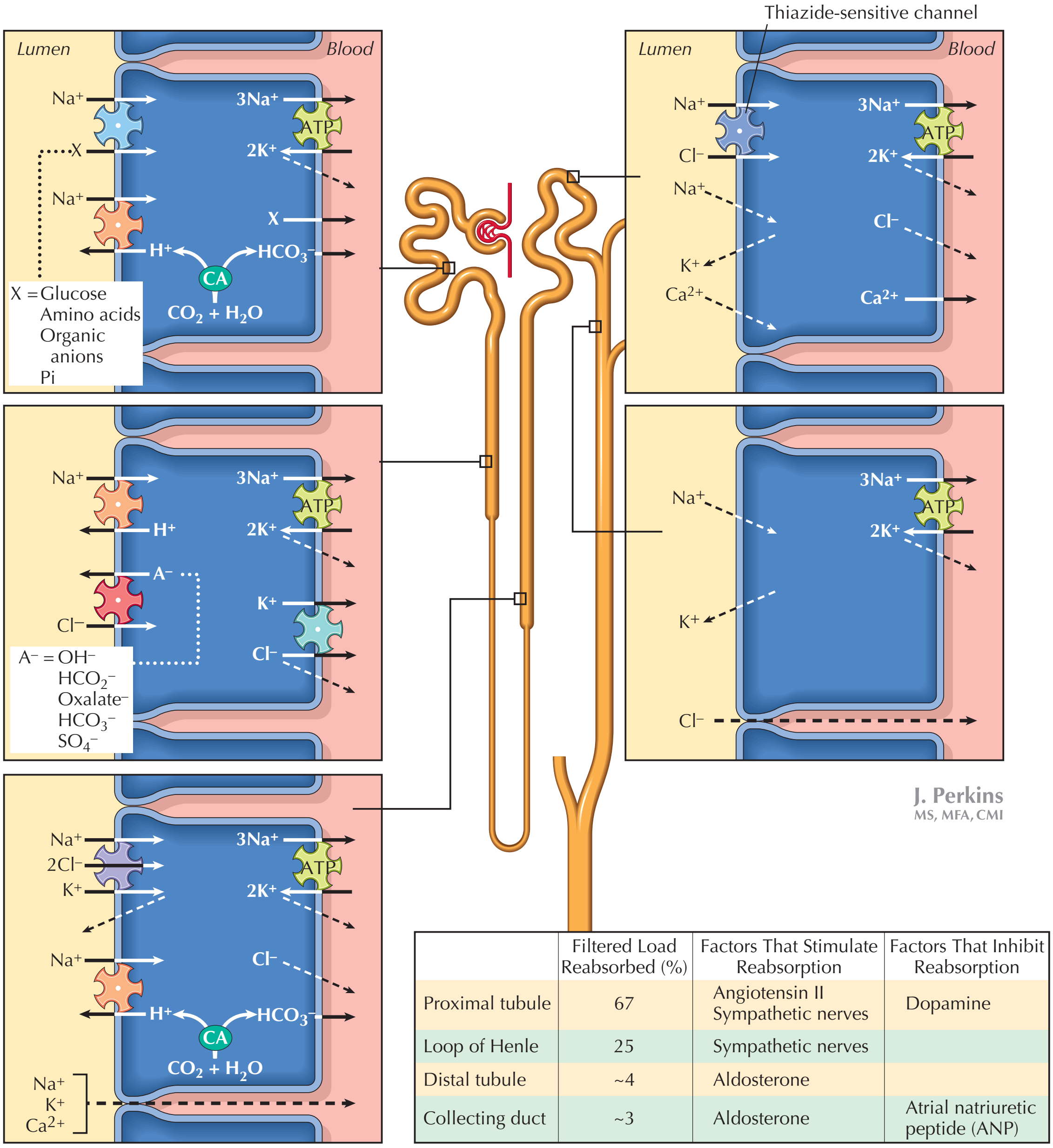
In general, of the total filtrate coming into the nephrons, the proximal tubule reabsorbs:

- 65% to 70% of the Na^+ and H_2O
- 80% to 85% of the K^+
- 65% of the Cl^-
- 75% to 80% of the phosphate
- 100% of the glucose
- 100% of the amino acids

Following this bulk reabsorption, "fine-tuning" of reabsorption occurs in subsequent segments of the nephron.

Approximately 65% to 70% of the water in tubular fluid is reabsorbed from the proximal tubule back into the peritubular capillaries, primarily following sodium reabsorption. The filtered load (FL) of sodium through the glomeruli is high (~25,000 mEq/day), and to maintain body fluid homeostasis, greater than 99% of the FL_{Na} must be reabsorbed back into the blood. This is accomplished by apical secondary active transport of sodium down a concentration gradient established by the basolateral **Na^+/K^+ ATPase** pumps. Figure 17.1 illustrates the primary sites and transporters for sodium reabsorption along different segments of the nephron.

- *Proximal convoluted tubule (S1 and S2 segments):* Bulk flow occurs by secondary active sodium cotransport with several substances including **glucose, amino acids, phosphate, and organic acids.** The proximal tubule also has **Na^+/H^+ antiporters,** which allow H^+ secretion into the proximal renal tubular fluid.
- *Proximal straight tubule (S3 segment):* **Na^+/H^+ antiporters** continue to reabsorb sodium and secrete H^+ into the tubular fluid. The reabsorption of sodium and fluid also provides the electrochemical gradient that facilitates **chloride** reabsorption. Cl^- concentration increases along the proximal tubule segments as water is reabsorbed. Chloride enters the cells in the S3 segment down its electrochemical gradient through antiporters, resulting in apical secretion of anions such as OH^-, HCO_3^-, SO_4^-, and oxalate. Cl^- reabsorption also occurs paracellularly, or between the cells. (The whole PT reabsorbs ~65% to 70% of FL_{Na}.)
- *Thin descending limb of Henle (tDLH):* This segment is impermeable to sodium and most other solutes but is permeable to water in the presence of antidiuretic hormone (ADH), and thus *concentrates* the tubular fluid (more on this in Chapter 18).



	Filtered Load Reabsorbed (%)	Factors That Stimulate Reabsorption	Factors That Inhibit Reabsorption
Proximal tubule	67	Angiotensin II Sympathetic nerves	Dopamine
Loop of Henle	25	Sympathetic nerves	
Distal tubule	~4	Aldosterone	
Collecting duct	~3	Aldosterone	Atrial natriuretic peptide (ANP)

Figure 17.1 Nephron Sites of Sodium Reabsorption Sodium reabsorption is critical for proper fluid and electrolyte homeostasis. More than 99% of the filtered load is reabsorbed through a variety of transport mechanisms. The gradient for sodium transport into the cells is maintained by basolateral Na^+/K^+ ATPase pumps.

- *Thick ascending limb of Henle (TALH):* This segment is impermeable to water, but specialized apical **Na^+-K^+-$2Cl^-$ cotransporters** facilitate reabsorption of electrolytes and *dilution* of the tubular fluid entering the distal tubule. These transporters are the targets for **loop diuretics** such as **furosemide** and **bumetanide.** In addition, there is a backleak of K^+ out of the cells into the lumen, creating a lumen-positive transepithelial potential difference (compared with interstitial fluid). This allows paracellular movement of cations (Ca^{2+}, Mg^{2+}, Na^+, K^+) out of the tubular lumen. In addition to the Na^+-K^+-$2Cl^-$ cotransporter, there are also **Na^+/H^+ antiporters,** which reabsorb Na^+ and secrete H^+ into the tubule. (TALH reabsorbs ~20% to 25% of FL_{Na}.)
- *Distal tubule (DT):* The early DT has **Na^+-Cl^- cotransporters** that can be inhibited by **thiazide diuretics.** The late DT has Na^+ (and K^+) channels that are increased by the hormone **aldosterone,** resulting in greater Na^+ and

water reabsorption. This aldosterone-sensitive epithelial sodium channel (ENaC) is blocked by amiloride, which is a potassium-sparing diuretic (discussed later). Aldosterone also responds to elevated plasma K^+, and increases distal and collecting tubule secretion of K^+. (DT reabsorbs ~4% of FL_{Na}.)

- *Collecting tubule:* Like the late DT, the collecting tubule has Na^+ (and K^+) channels that are increased by aldosterone. (CT reabsorbs ~3% of FL_{Na}.)

Glucose Transport

Because of the large FL_{Na}, the reabsorption of sodium is not a rate-limiting step in the reabsorption of other solutes. For many solutes, the rate-limiting step is the number of specific transporters available for the solute. Glucose is a good example of this concept. The sodium-glucose carriers have a high **transport maximum (TM),** and under normal conditions, the filtered load of glucose is low enough that the transporters can carry all of the solute back into the blood, leaving none in the tubular fluid and urine (Fig. 17.2). Thus, the renal clearance of glucose is normally zero.

However, if the FL of glucose is high, there may be too much glucose present in the tubular fluid and the carriers can become saturated. The renal threshold describes the point where the first nephrons exceed their TM, resulting in glucose in the urine **(glucosuria).** When the plasma glucose concentration (and hence the filtered load of glucose) is under the renal threshold for reabsorption, all of the glucose in tubular fluid will be reabsorbed (see Fig. 17.2). However, when it exceeds the threshold, the transporters are saturated (TM exceeded) and glucose appears in the urine.

The plasma concentration at which the renal threshold for glucose reabsorption is exceeded (and glucosuria is observed) is ~250 mg%. However, the calculated plasma threshold is 300 mg%. This difference between real and calculated values is explained by nephron heterogeneity (also called splay), whereby different nephron populations have higher and lower TMs for glucose. The average TM (for both kidneys) is the basis for the calculation of the threshold for plasma glucose levels (at which glucosuria occurs), despite the fact that some nephrons have a lower TM that will be exceeded when plasma glucose is over ~250 mg%.

This concept is important in diabetes mellitus, in which the inability to efficiently transport glucose into tissues leads to high plasma glucose concentrations. The fasting plasma glucose is much higher than normal in diabetes (greater than 130 mg% compared to 80 to 90 mg%), resulting in increased FL of glucose. With feeding, the plasma levels can easily exceed the TM of some nephrons, causing glucosuria. In addition, because glucose is an osmotic agent, the glucosuria will be associated with diuresis (loss of water through increased urine volume).

BICARBONATE HANDLING

Plasma **bicarbonate** is necessary for acid–base homeostasis. At normal whole body pH balance, 100% of the filtered bicarbonate (HCO_3^-) is effectively reabsorbed. However, this occurs *indirectly* through a process involving H^+ secretion (through cation exchange and active H^+ pumps). In the tubular lumen, filtered HCO_3^- and secreted H^+ form CO_2 and H_2O (a reaction catalyzed by brush border carbonic anhydrase, CA), which diffuse into the cell (Fig. 17.3). Once in the cell, the CO_2 and H_2O are converted back to carbonic acid (by intracellular CA); HCO_3^- is transported out of the cell via basolateral HCO_3^-/Cl^- exchangers or Na^+-HCO_3^- cotransporters, depending on the nephron segment. The H^+ generated from this process is secreted back into the tubular lumen and can be used to reabsorb more HCO_3^-, or can be buffered and excreted (see Chapter 20). This mechanism is present in three segments of the nephron, facilitating reabsorption of filtered bicarbonate in the PT (80% of filtered load), TALH (15%), and CD (5%).

Under normal conditions, the renal clearance of HCO_3^- is 0, meaning there is none in the urine. The regulation of bicarbonate handling is an integral part of acid–base homeostasis and will be discussed in Chapter 20.

POTASSIUM HANDLING

As with all of the major electrolytes, potassium balance is important to overall homeostasis, and dietary intake must be matched by urinary and fecal excretion. Plasma K^+ concentration must be maintained at relatively low levels (3 to 5 mEq/L) and is regulated by the kidneys. Potassium is pumped into cells (via Na^+/K^+ ATPase, which is stimulated by insulin and epinephrine), and the excess in the extracellular fluid (ECF) is excreted in urine. Figure 17.4 illustrates potassium handling through the nephron and the effects of dietary K^+ intake.

Potassium handling varies along the nephron:

- *Proximal tubule:* Potassium reabsorption occurs by **paracellular movement,** not by entry into the cells. Reabsorption initially occurs via **solvent drag,** initiated by water reabsorption. In the S2 and S3 segments, the positive potential of the tubular lumen allows (paracellular) potassium reabsorption by **diffusion** down the electrochemical gradient (this accounts for ~70% reabsorption of filtered potassium).
- *Thick ascending limb of Henle:* The Na^+-K^+-$2Cl^-$ cotransporters in the TALH use the sodium and chloride gradients to facilitate transport of K^+ (~20% of filtered potassium).
- *Late distal tubules:* Potassium can be secreted into the DTs via aldosterone-sensitive K^+ channels.
- *Collecting ducts:* Potassium is *secreted* into the collecting ducts through aldosterone-sensitive apical K^+ channels

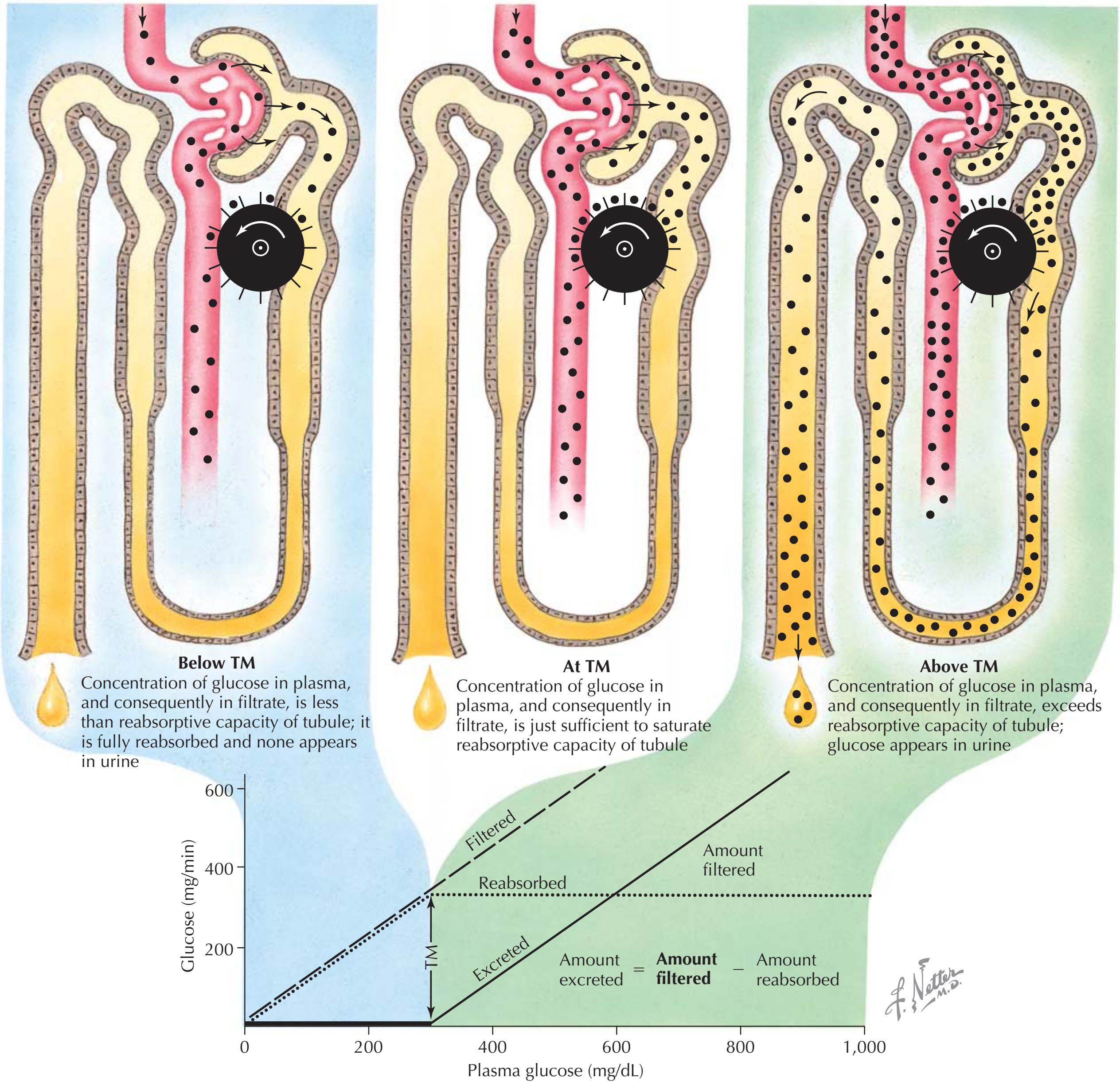


Figure 17.2 **Renal Handling of Glucose** Glucose is freely filtered at the glomerulus and is 100% reabsorbed in the proximal tubules by sodium-glucose cotransporters. However, if blood glucose levels become elevated, as in diabetes, the maximal tubular reabsorption rate (TM) is exceeded, and glucose appears in the urine *(far right panel).*

in principal cells. K^+ is also secreted into the collecting ducts by α-intercalated cells, in exchange for H^+. Under normal conditions there is a net secretion of K^+. Net reabsorption can occur during dietary K^+ depletion.

Renal potassium handling is influenced by the following:

- *Dietary potassium intake:* Increased intestinal K^+ absorption elevates plasma K^+ concentration. The

Loop diuretics, such as furosemide (Lasix) and bumetanide, inhibit the Na^+-K^+-$2Cl^-$ cotransporters, causing natriuresis/diuresis, which is beneficial in controlling hypertension. Extended use can cause urinary K^+ loss, and plasma K^+ must be monitored. Potassium-sparing diuretics, such as thiazides, target the distal tubule Na^+-Cl^- cotransporters and control potassium losses.

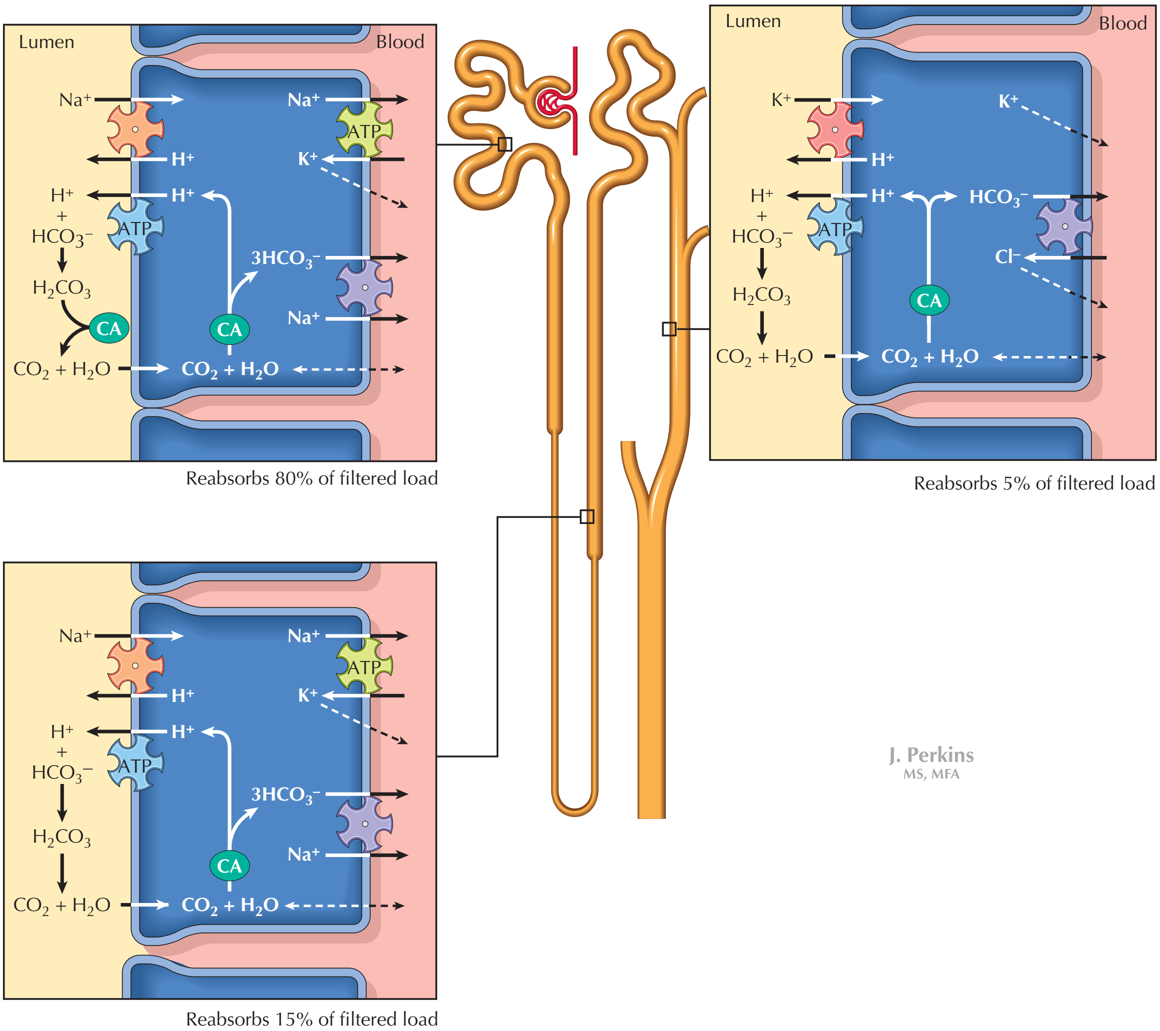


Figure 17.3 Renal HCO_3^- Reabsorption Bicarbonate is freely filtered at the glomerulus and is reabsorbed along the nephron through a process involving secretion of H^+. Under normal conditions, 100% of the filtered bicarbonate is reabsorbed.

mineralocorticoid aldosterone increases basolateral Na^+/K^+ ATPase activity, pumping more K^+ into the cells (see Chapter 19). K^+ is then secreted into the collecting ducts through apical channels in the principal cells. When dietary K^+ intake is low, K^+ secretion from the principal cells is inhibited, and K^+ reabsorption from the collecting duct α-intercalated cells predominates.

- *Plasma volume:* In addition to responding to increased plasma K^+ concentrations, aldosterone is also released in response to decreased plasma volume, as part of the renin-angiotensin-aldosterone system (RAAS). Aldosterone increases the Na^+/K^+ ATPase, Na^+/H^+ antiporters, and Na^+-Cl^- cotransporters in the late distal tubules and CDs, independent of plasma potassium levels. This increases K^+ secretion into the renal tubules, as stated earlier.
- *Acid–base status:* To compensate for acidosis, H^+ can be secreted into the collecting ducts from the principal cells while K^+ is reabsorbed. Conversely, during alkalosis, H^+ will be retained, and K^+ will be secreted from CD α-intercalated cells (see Chapter 20).
- *Tubular fluid flow rate:* When tubular fluid flow is high, the concentration gradient for K^+ (from collecting duct cell to the lumen) is high, and K^+ secretion will increase.

CALCIUM AND PHOSPHATE TRANSPORT

Calcium and phosphate are important during fetal and childhood development for bone and tissue growth, and continue to be important in the adult for bone health. The kidneys control plasma levels of calcium and phosphate by altering